

A General User-Customizable Predict+Optimize Framework

(Supplementary)

23.10.06

Kangyeon Kim

Contents

Preliminaries

Section 2: A General Controllable Predict+Optimize Framework

Section 3: Dealing Constraints: Chance-Constrained Method

Section 4: Training with Internal Test Cases

Section 5: Construction of the Loss function for Training

Section 6: Calculation of gradient (brief)

Section 7: Application 1: Stock trading

Section 8: Application 2: Safety gym

Preliminaries

Preliminaries

1. Compactness

[Theorem 0.0] (Heine-Borel Theorem)

$\mathbb{R}$ 에서의 닫힌구간 $I = [a, b]$ 에 대하여 $\{G_i\}$ 를 $I \subset \cup_i G_i$ 를 만족시키는 열린 집합들의 모임이라고 하자. (즉, $\{G_i\}$ 는 집합 A 의 열린 덮개이다.) 이때 $\{G_i\}$ 의 원소 $G_{i_1}, G_{i_2}, \dots, G_{i_m}$ 가 존재하여

$$I \subset G_{i_1} \cup G_{i_2} \cup \dots \cup G_{i_m}$$

을 만족시킨다.

[Definition 0.1]

위상공간 X 의 부분집합 A 의 임의의 열린 덮개가 항상 유한인 부분덮개를 가지면 집합 A 를 콤팩트(compact)하다고 한다.

Theorem 2.41 Let $\{E \in \mathbb{R}^k\}$. The following properties are equivalent:

- (a) E is closed and bounded.
- (b) E is compact.

Preliminaries

1. Compactness

1) compact $\Leftrightarrow$ closed and bounded

Preliminaries

2. Continuity

Def 3.2.1) [Continuous function]

(1) Let E be a subset of $\mathbb{R}$ and f is real-valued function with domain E . The function f is continuous point at $a \in E$ if given $\varepsilon > 0$, there exists $\delta > 0$ s.t $|f(x) - f(a)| < \varepsilon$, for all $x \in E$ with $|x - a| < \delta$, which denoted $\lim_{x \rightarrow a} f(x) = f(a)$.

(2) The function f is continuous on E if and only if f is continuous at every point E .

Def 3.3.1) [Uniform Continuity]

Let E be a subset of $\mathbb{R}$ and f is real-valued function with domain E . The function f is uniform continuous on E if given $\varepsilon > 0$, there exists $\delta > 0$ s.t,

$$|f(x) - f(y)| < \varepsilon, \text{ for all } x, y \in E, \text{ with } |x - y| < \delta$$

Preliminaries

3. Theorem

Theorem 1.9. *Let f be a map from a topological space on X to a topological space on Y . Then, the following statements are equivalent:*

1. f is continuous.
2. Inverse image of every closed set of Y is closed in X .

Theorem 6.2.9: Continuity and Uniform Continuity

If f is uniformly continuous in a domain D , then f is continuous in D .

If f is continuous on a compact domain D , then f is uniformly continuous in D .

정리 1 (극값 정리 또는 최대-최소 정리)

집합 $E \subseteq \mathbb{R}^n$ 를 긴밀 집합(compact set)이라 하고 함수 $f: E \rightarrow \mathbb{R}$ 가 연속함수(continuous function)라 하자. 그러면 함수 f 는 집합 E 안에서 언제나 최댓값(maximum)과 최솟값(minimum)을 갖는다. 다시 말해, 임의의 $x \in E$ 에 대하여 $f(x^*) \leq f(x) \leq f(y^*)$ 를 만족하는 $x^*, y^* \in E$ 가 존재한다.

Preliminaries

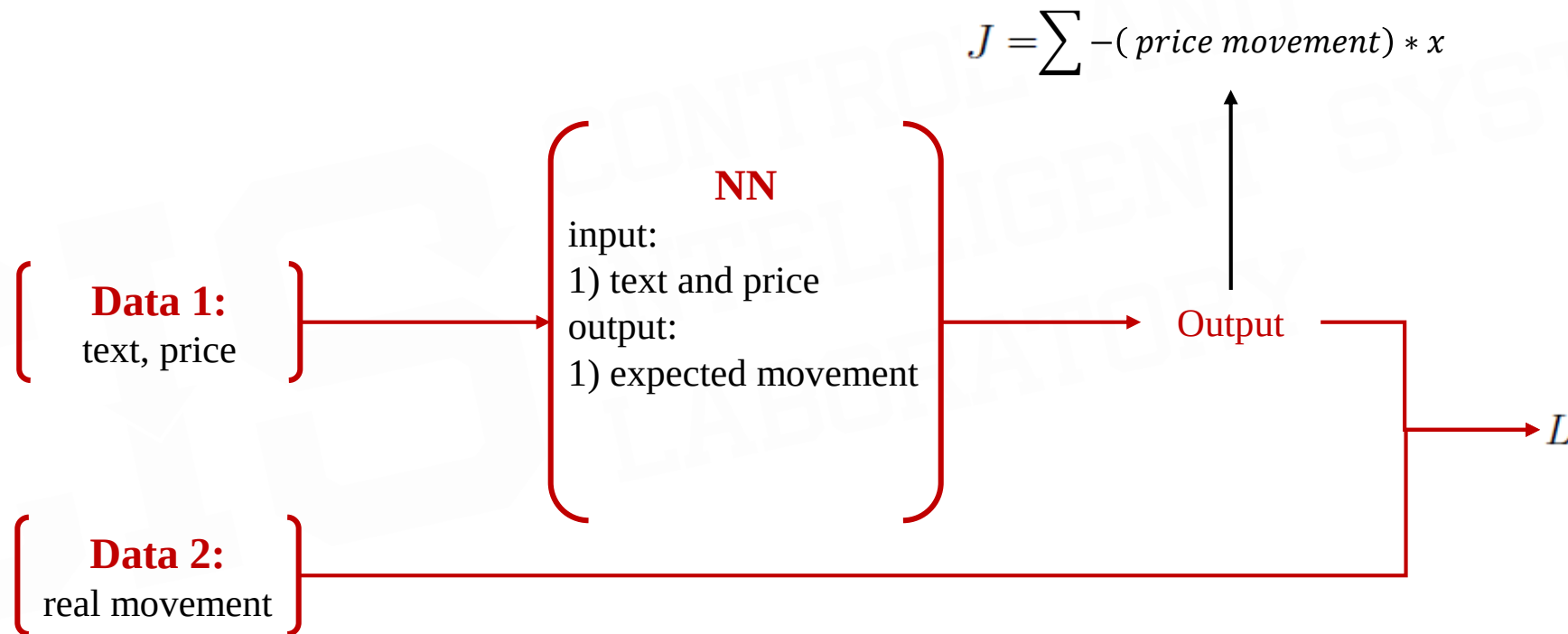
3. Theorem

- 1) continuous + inverse image of closed set = closed set
- 2) continuous + compact domain = uniformly continuous
- 3) continuous + compact domain = max, min exist

Preliminaries

4. Application shortcut (Predict+Optimize Framework)

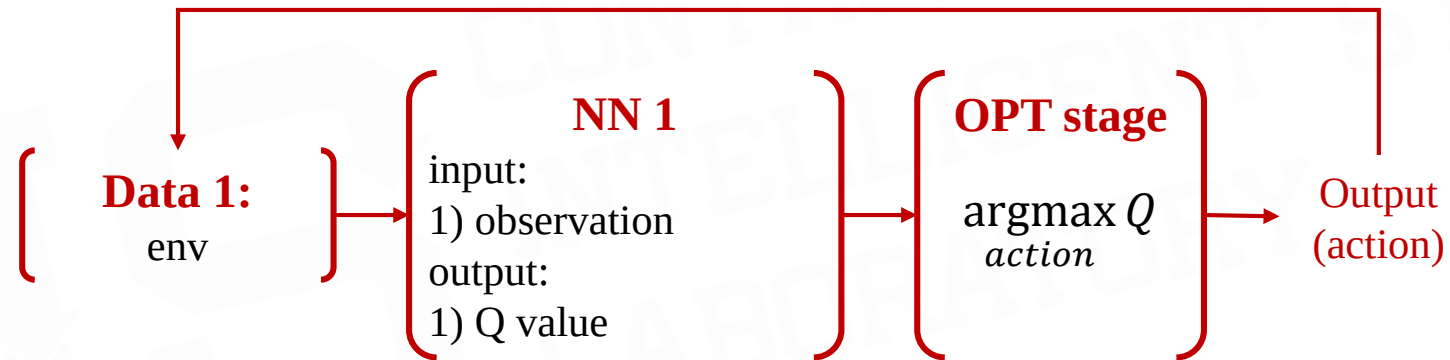
1) Application 1: Stock trading



Preliminaries

4. Application shortcut (Predict+Optimize Framework)

2) Application 2: Safety gym (RL, DQN)



Section 2: A General Controllable Predict+Optimize Framework

Section 2: A General Controllable Predict+Optimize Framework

1. Problem Setup

1) Goal: minimize $J(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r})$

s.t. $c_i(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r}) \geq 0, \quad i = 1, \dots, n_c$

2) Notation:

- continuous variable: $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^{n_x}$
- discrete variable: $\mathbf{z} \in \mathcal{Z} \subset N_0^{n_z}$
- user-given parameters: $\mathbf{r} \in \mathcal{R} \subset \mathbb{R}^{n_{rc}} \times N_0^{n_{rd}}$
- environment: $\boldsymbol{\theta} (\in \Theta)$

Section 2: A General Controllable Predict+Optimize Framework

1. Problem Setup

1) Goal: minimize $J(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r})$

s.t. $c_i(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r}) \geq 0, \quad i = 1, \dots, n_c$

2) Notation:

- continuous variable: $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^{n_x}$

- discrete variable: $\mathbf{z} \in \mathcal{Z} \subset N_0^{n_z}$

- user-given parameters: $\mathbf{r} \in \mathcal{R} \subset \mathbb{R}^{n_{rc}} \times N_0^{n_{rd}}$

- environment: $\boldsymbol{\theta} \in \Theta$

explain 1: to deal with general optimization problem.

Section 2: A General Controllable Predict+Optimize Framework

1. Problem Setup

1) Goal: minimize $J(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r})$

s.t. $c_i(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r}) \geq 0, \quad i = 1, \dots, n_c$

2) Notation:

- continuous variable: $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^{n_x}$

- discrete variable: $\mathbf{z} \in \mathcal{Z} \subset N_0^{n_z}$

- user-given parameters: $\mathbf{r} \in \mathcal{R} \subset \mathbb{R}^{n_{rc}} \times N_0^{n_{rd}}$

- environment: $\boldsymbol{\theta} \in \Theta$

explain 2: for customization.

be regarded as ‘knob’ for users.

Section 2: A General Controllable Predict+Optimize Framework

1. Problem Setup

1) Goal: minimize $J(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r})$

s.t. $c_i(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r}) \geq 0, \quad i = 1, \dots, n_c$

2) Notation:

- continuous variable: $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^{n_x}$

- discrete variable: $\mathbf{z} \in \mathcal{Z} \subset N_0^{n_z}$

- user-given parameters: $\mathbf{r} \in \mathcal{R} \subset \mathbb{R}^{n_{rc}} \times N_0^{n_{rd}}$

- environment: $\boldsymbol{\theta} (\in \Theta)$

issue 1: we do not have full information about environment.

→ *Measurement*: $\mathbf{y}$ that follows the function $Samp(\boldsymbol{\theta})$

Section 2: A General Controllable Predict+Optimize Framework

1. Problem Setup

- 1) Goal: minimize $J(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r})$ **issue 2: can be unknown.**
even if it is known, it is a function of the environment.
s.t. $c_i(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r}) \geq 0, \quad i = 1, \dots, n_c$

2) Notation:

- continuous variable: $\mathbf{x} \in \mathcal{X} \subset R^{n_x}$
- discrete variable: $\mathbf{z} \in \mathcal{Z} \subset N_0^{n_z}$
- user-given parameters: $\mathbf{r} \in \mathcal{R} \subset R^{n_{rc}} \times N_0^{n_{rd}}$
- environment: $\boldsymbol{\theta} (\in \Theta)$

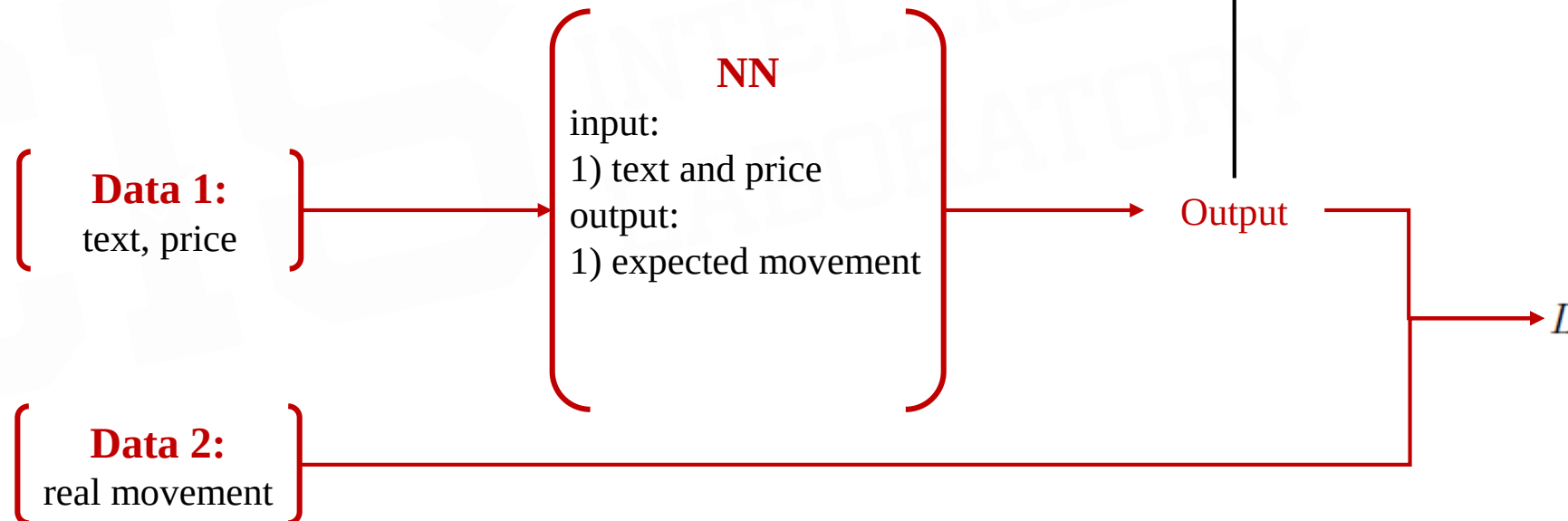
Section 2: A General Controllable Predict+Optimize Framework

1. Problem Setup

1) Goal: minimize $J(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r})$ example 1: application 1 & 2

s.t. $c_i(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r}) \geq 0, \quad i = 1, \dots, n_c$

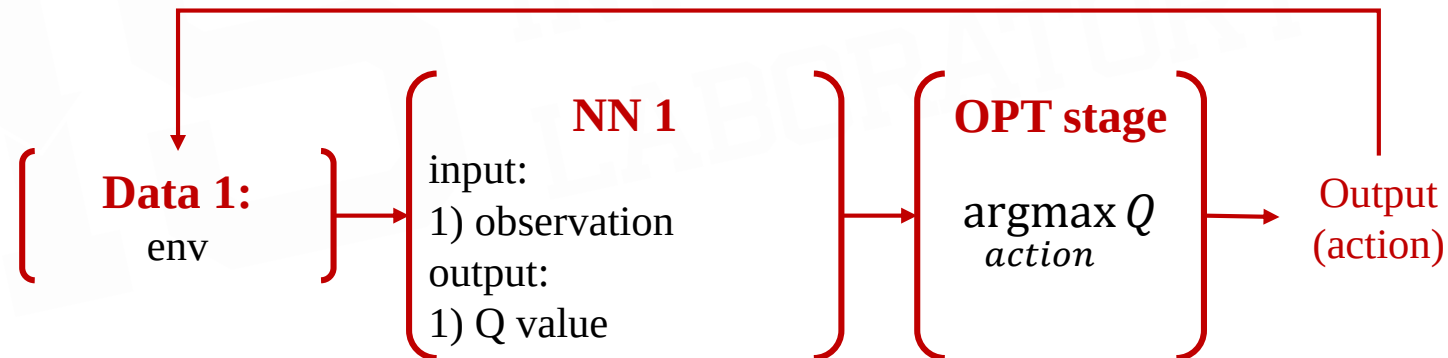
$$J = \sum -(\text{price movement}) * x$$



Section 2: A General Controllable Predict+Optimize Framework

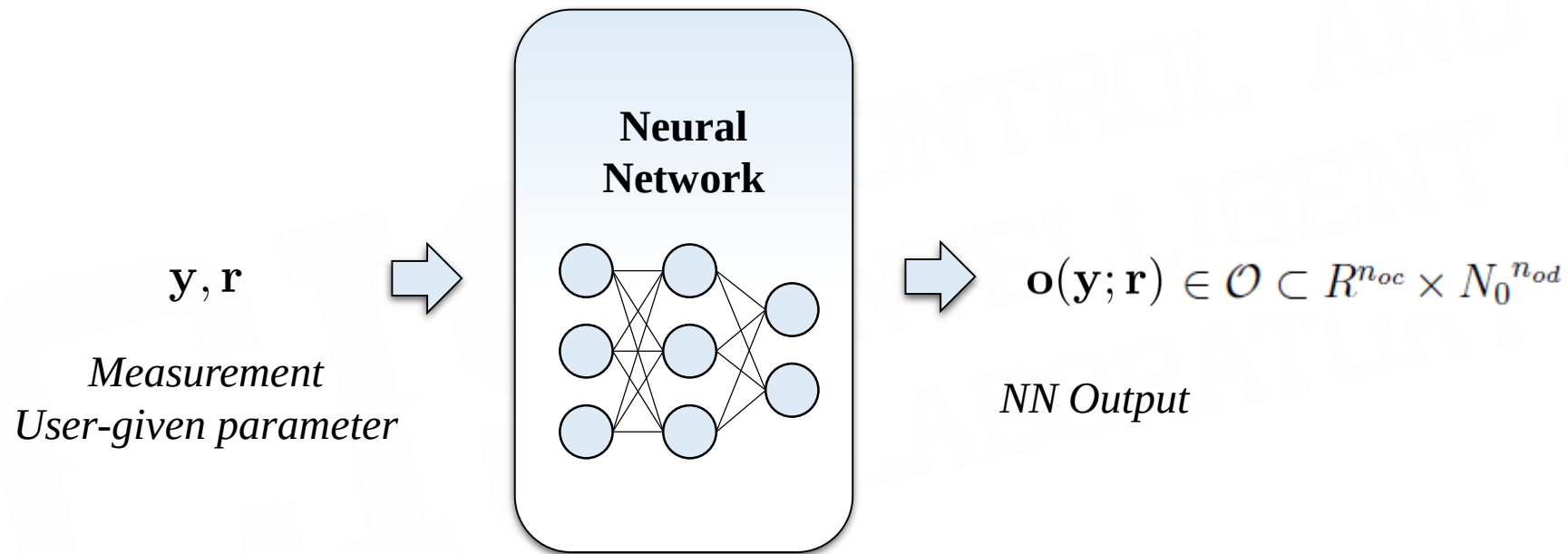
1. Problem Setup

- 1) Goal: minimize $J(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r})$ example 1: application 1 & 2
s.t. $c_i(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r}) \geq 0, \quad i = 1, \dots, n_c$



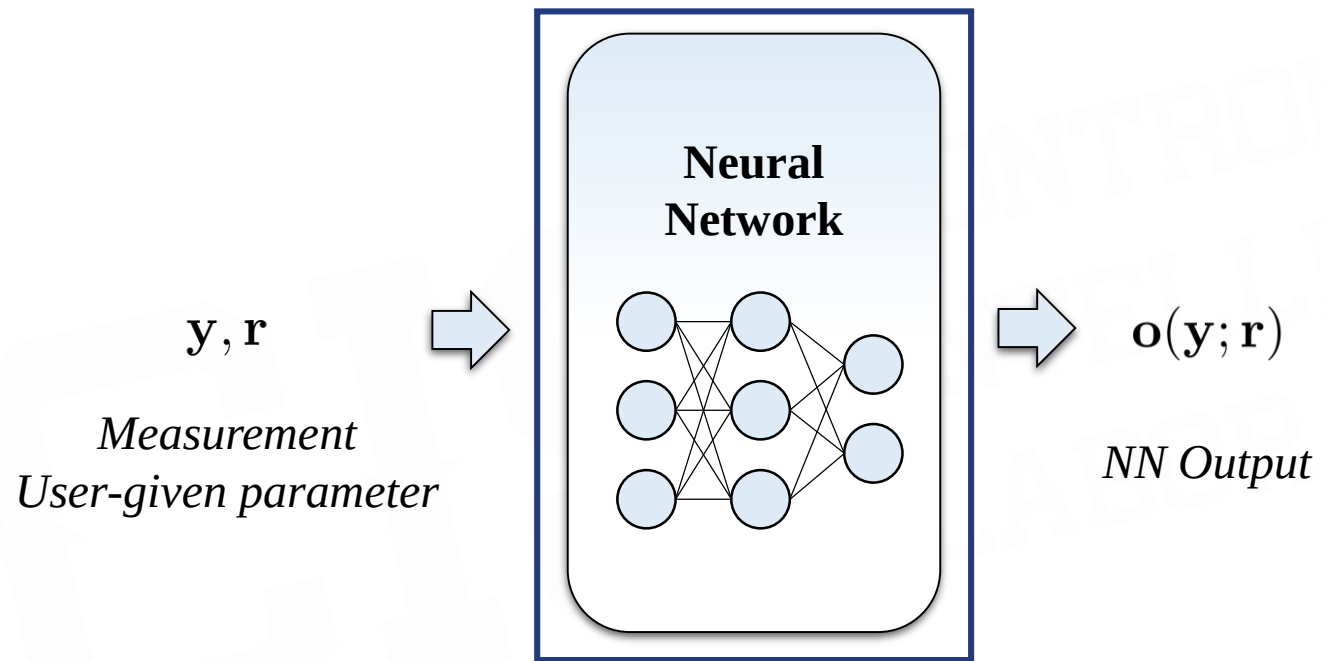
Section 2: A General Controllable Predict+Optimize Framework

2. Predict+Optimize Framework Formulation



Section 2: A General Controllable Predict+Optimize Framework

2. Predict+Optimize Framework Formulation

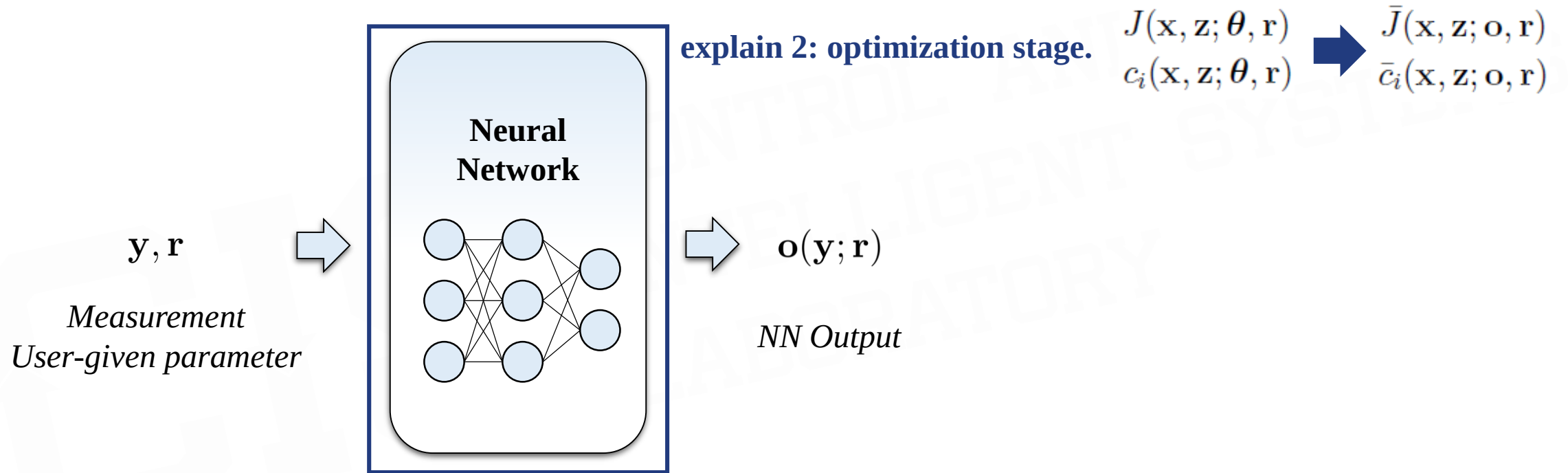


explain 1: NN processes measurements to obtain tailored output to achieve the best performance under parameter.

→ *Perception: measurement + postprocessing(NN)*

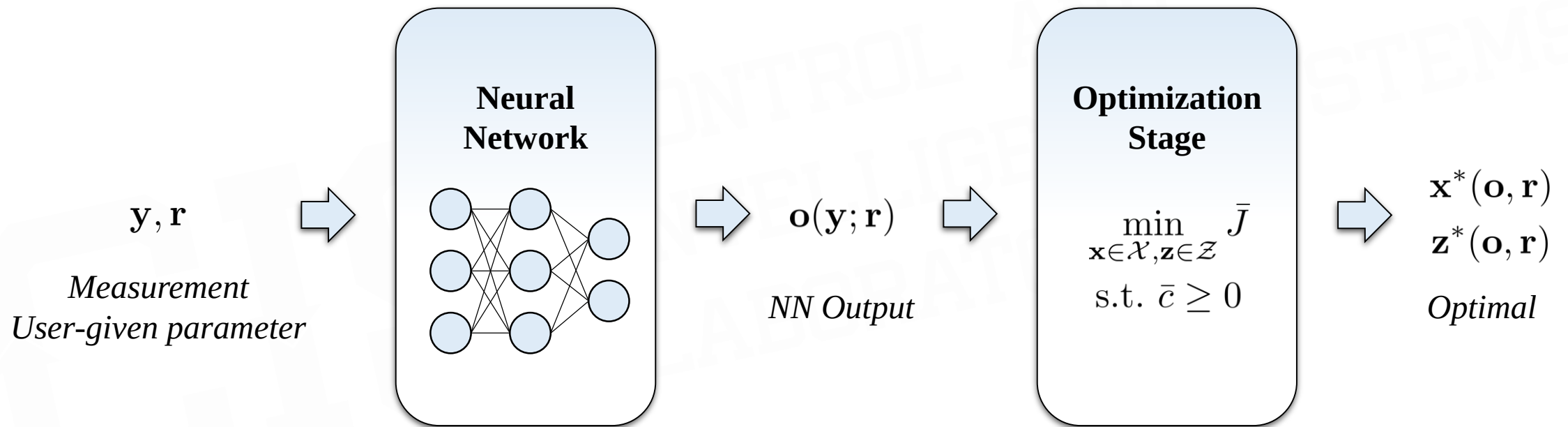
Section 2: A General Controllable Predict+Optimize Framework

2. Predict+Optimize Framework Formulation



Section 2: A General Controllable Predict+Optimize Framework

2. Predict+Optimize Framework Formulation



Section 3: Dealing Constraints

Section 3: Dealing Constraints: Chance-Constrained Method

1. Constraints: How to deal?

1) Motivation:

- to ensure safety. to guarantee constraint satisfaction.

2) Constraints assumption

- do not depend on environment: c_i for $i = 1, \dots, n_{cg}$
- depend on environment: c_i for $i = n_{cg} + 1, \dots, n_c$

Section 3: Dealing Constraints: Chance-Constrained Method

1. Constraints: How to deal?

1) Motivation:

- to ensure safety. to guarantee constraint satisfaction.

explain 1: easy to deal.

2) Constraints assumption

can be directly considered in the decision making stage.

- do not depend on environment: c_i for $i = 1, \dots, n_{cg}$

- depend on environment: c_i for $i = n_{cg} + 1, \dots, n_c$

Section 3: Dealing Constraints: Chance-Constrained Method

1. Constraints: How to deal?

1) Motivation:

- to ensure safety. to guarantee constraint satisfaction.

2) Constraints assumption

- do not depend on environment: c_i for $i = 1, \dots, n_{cg}$

- depend on environment: c_i for $i = n_{cg} + 1, \dots, n_c$

**issue 1: we should obtain the information about the environment.
cannot be directly considered in the decision making stage.**

- *Obtaining the distribution of environment, and deal constraint with probability.*
- *'Internal test'*

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

1) Notation

- ‘ncr’: not including information needed in some constraints.
- ‘cr’: including information needed in some constraints.

$$\begin{aligned}\theta &= (\theta_{ncr}(\in \Theta_{ncr}), \theta_{cr}(\in \Theta_{cr})) \\ y &= (y_{ncr}, y_{cr}) \\ o &= (o_{ncr}(\in \mathcal{O}_{ncr}), o_{cr}(\in \mathcal{O}_{cr}))\end{aligned}$$

2) Remark

- environment-dependent constraints(c_i for $i = n_{cg} + 1, \dots, n_c$) are only function of θ_{cr} .
- we only deal with the case that both Θ_{cr} and $\mathcal{O}_{cr}$ are finite.

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

1) Notation

- ‘ ncr ’: not including information needed in some constraints.
- ‘ cr ’: including information needed in some constraints.

$$\begin{aligned}\theta &= (\theta_{ncr}(\in \Theta_{ncr}), \theta_{cr}(\in \Theta_{cr})) \\ y &= (y_{ncr}, y_{cr}) \\ o &= (o_{ncr}(\in \mathcal{O}_{ncr}), o_{cr}(\in \mathcal{O}_{cr}))\end{aligned}$$

2) Remark

- environment-dependent constraints(c_i for $i = n_{cg} + 1, \dots, n_c$) are only function of θ_{cr} .

- we only deal with the case that both Θ_{cr} and $\mathcal{O}_{cr}$ are finite.

explain 1: both θ_{cr} and o_{cr} do not have any continuous part.

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

$$\forall i, Pr(c_i(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}_{cr}, \mathbf{r}) < 0) \leq r_{t,i}$$

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

explain 2: environment-dependent constraint

$$\forall i, Pr\left(c_i(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}_{cr}, \mathbf{r}) < 0\right) \leq r_{t,i}$$

explain 3: including in user parameter.

$\rightarrow (r_{t,n_{cg}+1}, \dots, r_{t,n_c})$

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

$$\forall i, Pr(c_i(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}_{cr}, \mathbf{r}) < 0) \leq r_{t,i}$$

issue 1: how to calculate the left hand side?

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

4) Notation

- real environment: superscript r
- internal test case: superscript t_i
- the number of internal test cases: N_t

5) Measurement and Output

$$\mathbf{y} = (\mathbf{y}^r, \mathbf{y}_{cr}^{t_1}, \dots, \mathbf{y}_{cr}^{t_{N_t}}), \mathbf{y}^r \sim \text{Samp}(\boldsymbol{\theta}^r), \mathbf{y}_{cr}^{t_i} \sim \text{Samp}_{cr}(\boldsymbol{\theta}_{cr}^{t_i})$$

$$\mathbf{o}((\mathbf{y}^r, \mathbf{y}_{cr}^{t_1}, \dots, \mathbf{y}_{cr}^{t_{N_t}}); \mathbf{r}) := (\mathbf{o}^r, \mathbf{o}_{cr}^{t_1}, \dots, \mathbf{o}_{cr}^{t_{N_t}}) := (\mathbf{o}(\mathbf{y}^r; \mathbf{r}), \mathbf{o}_{cr}(\mathbf{y}_{cr}^{t_1}; \mathbf{r}), \dots, \mathbf{o}_{cr}(\mathbf{y}_{cr}^{t_{N_t}}; \mathbf{r}))$$

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

4) Notation

- real environment: superscript r
- internal test case: superscript t_i
- the number of internal test cases: N_t

5) Measurement and Output

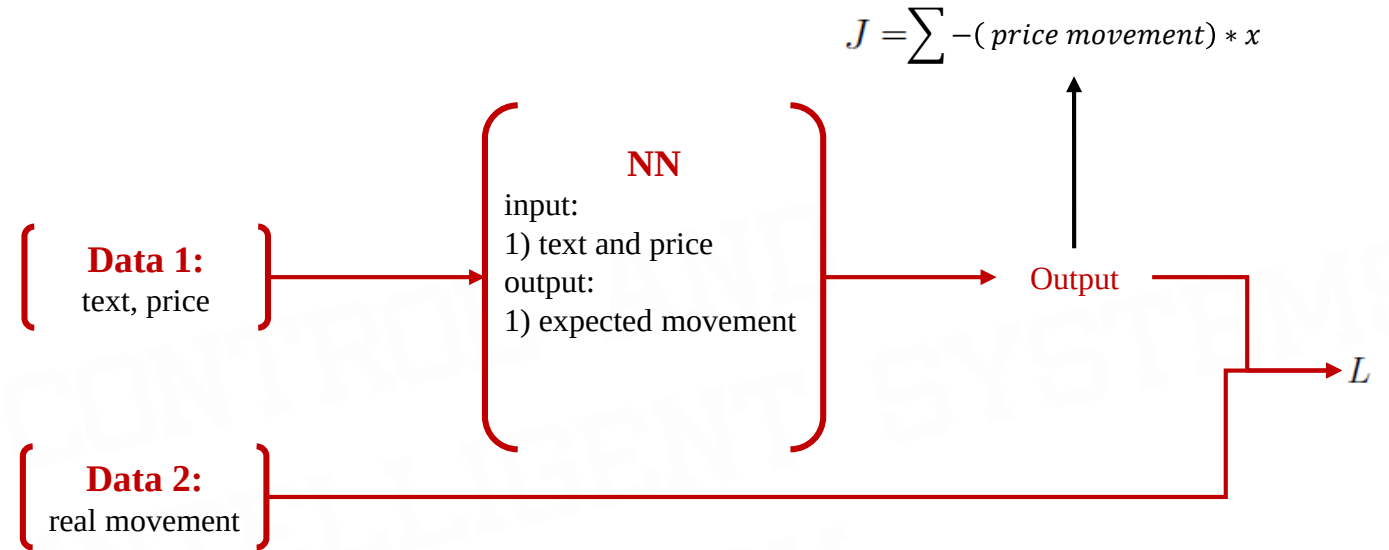
explain 4: measurement of the fixed environment(cr) of internal test cases.

$$\mathbf{y} = (\mathbf{y}^r, \boxed{\mathbf{y}_{cr}^{t_1}, \dots, \mathbf{y}_{cr}^{t_{N_t}}}), \mathbf{y}^r \sim \text{Samp}(\boldsymbol{\theta}^r), \mathbf{y}_{cr}^{t_i} \sim \text{Samp}_{cr}(\boldsymbol{\theta}_{cr}^{t_i})$$

$$\mathbf{o}((\mathbf{y}^r, \mathbf{y}_{cr}^{t_1}, \dots, \mathbf{y}_{cr}^{t_{N_t}}); \mathbf{r}) := (\mathbf{o}^r, \mathbf{o}_{cr}^{t_1}, \dots, \mathbf{o}_{cr}^{t_{N_t}}) := (\mathbf{o}(\mathbf{y}^r; \mathbf{r}), \mathbf{o}_{cr}(\mathbf{y}_{cr}^{t_1}; \mathbf{r}), \dots, \mathbf{o}_{cr}(\mathbf{y}_{cr}^{t_{N_t}}; \mathbf{r}))$$

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method



5) Measurement and Output

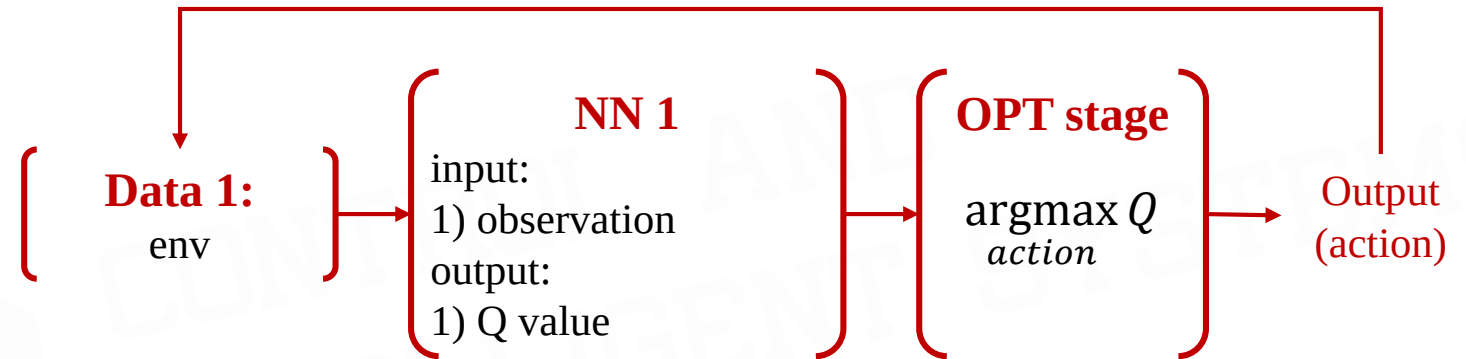
issue 2: application 1 & 2

$$\mathbf{y} = (\mathbf{y}^r, \boxed{\mathbf{y}_{cr}^{t_1}, \dots, \mathbf{y}_{cr}^{t_{N_t}}}), \mathbf{y}^r \sim \text{Samp}(\boldsymbol{\theta}^r), \mathbf{y}_{cr}^{t_i} \sim \text{Samp}_{cr}(\boldsymbol{\theta}_{cr}^{t_i})$$

$$\mathbf{o}((\mathbf{y}^r, \mathbf{y}_{cr}^{t_1}, \dots, \mathbf{y}_{cr}^{t_{N_t}}); \mathbf{r}) := (\mathbf{o}^r, \mathbf{o}_{cr}^{t_1}, \dots, \mathbf{o}_{cr}^{t_{N_t}}) := (\mathbf{o}(\mathbf{y}^r; \mathbf{r}), \mathbf{o}_{cr}(\mathbf{y}_{cr}^{t_1}; \mathbf{r}), \dots, \mathbf{o}_{cr}(\mathbf{y}_{cr}^{t_{N_t}}; \mathbf{r}))$$

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method



5) Measurement and Output

issue 2: application 1 & 2

$$\mathbf{y} = (\mathbf{y}^r, \boxed{\mathbf{y}_{cr}^{t_1}, \dots, \mathbf{y}_{cr}^{t_{N_t}}}), \mathbf{y}^r \sim \text{Samp}(\boldsymbol{\theta}^r), \mathbf{y}_{cr}^{t_i} \sim \text{Samp}_{cr}(\boldsymbol{\theta}_{cr}^{t_i})$$

$$\mathbf{o}((\mathbf{y}^r, \mathbf{y}_{cr}^{t_1}, \dots, \mathbf{y}_{cr}^{t_{N_t}}); \mathbf{r}) := (\mathbf{o}^r, \mathbf{o}_{cr}^{t_1}, \dots, \mathbf{o}_{cr}^{t_{N_t}}) := (\mathbf{o}(\mathbf{y}^r; \mathbf{r}), \mathbf{o}_{cr}(\mathbf{y}_{cr}^{t_1}; \mathbf{r}), \dots, \mathbf{o}_{cr}(\mathbf{y}_{cr}^{t_{N_t}}; \mathbf{r}))$$

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

4) Notation

- real environment: superscript r
- internal test case: superscript t_i
- the number of internal test cases: N_t

5) Measurement and Output

$$\mathbf{y} = (\mathbf{y}^r, \mathbf{y}_{cr}^{t_1}, \dots, \mathbf{y}_{cr}^{t_{N_t}}), \mathbf{y}^r \sim \text{Samp}(\boldsymbol{\theta}^r), \mathbf{y}_{cr}^{t_i} \sim \text{Samp}_{cr}(\boldsymbol{\theta}_{cr}^{t_i})$$

$$\mathbf{o}((\mathbf{y}^r, \mathbf{y}_{cr}^{t_1}, \dots, \mathbf{y}_{cr}^{t_{N_t}}); \mathbf{r}) := (\mathbf{o}^r, \mathbf{o}_{cr}^{t_1}, \dots, \mathbf{o}_{cr}^{t_{N_t}}) := (\mathbf{o}(\mathbf{y}^r; \mathbf{r}), \mathbf{o}_{cr}(\mathbf{y}_{cr}^{t_1}; \mathbf{r}), \dots, \mathbf{o}_{cr}(\mathbf{y}_{cr}^{t_{N_t}}; \mathbf{r}))$$

explain 5: forward all measurement results parallely into the neural network.

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

6) Notation

- constraint related env set (finite): $\Theta_{cr} = \{\bar{\theta}_1, \dots, \bar{\theta}_{N_e}\}$
- constraint related output set (finite): $\mathcal{O}_{cr} = \{\bar{o}_1, \dots, \bar{o}_{N_o}\}$
- number of internal test cases for each env: $N_{\theta_{cr}=\bar{\theta}_i} = \sum_{k=1}^{N_t} \mathbf{1}(\theta_{cr}^{t_k}, \bar{\theta}_i)$
- number of internal test cases for each env and each output: $N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j} = \sum_{k=1}^{N_t} \mathbf{1}(\theta_{cr}^{t_k}, \bar{\theta}_i) \mathbf{1}(o_{cr}^{t_k}, \bar{o}_j)$

7) Probability for each output

$$\underline{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i)} \leq p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i) \leq \overline{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i)}$$

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

6) Notation

- constraint related env set (finite): $\Theta_{cr} = \{\bar{\theta}_1, \dots, \bar{\theta}_{N_e}\}$
- constraint related output set (finite): $\mathcal{O}_{cr} = \{\bar{o}_1, \dots, \bar{o}_{N_o}\}$
- number of internal test cases for each env: $N_{\theta_{cr}=\bar{\theta}_i} = \sum_{k=1}^{N_t} \mathbf{1}(\theta_{cr}^{t_k}, \bar{\theta}_i)$
- number of internal test cases for each env and each output: $N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j} = \sum_{k=1}^{N_t} \mathbf{1}(\theta_{cr}^{t_k}, \bar{\theta}_i) \mathbf{1}(o_{cr}^{t_k}, \bar{o}_j)$

7) Probability for each output

$$\underline{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i)} \leq \boxed{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i)} \leq \overline{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i)}$$

explain 6: $\frac{N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j}}{N_{\theta_{cr}=\bar{\theta}_i}}$

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

6) Notation

- constraint related env set (finite): $\Theta_{cr} = \{\bar{\theta}_1, \dots, \bar{\theta}_{N_e}\}$
- constraint related output set (finite): $\mathcal{O}_{cr} = \{\bar{o}_1, \dots, \bar{o}_{N_o}\}$
- number of internal test cases for each env: $N_{\theta_{cr}=\bar{\theta}_i} = \sum_{k=1}^{N_t} \mathbf{1}(\theta_{cr}^{t_k}, \bar{\theta}_i)$
- number of internal test cases for each env and each output: $N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j} = \sum_{k=1}^{N_t} \mathbf{1}(\theta_{cr}^{t_k}, \bar{\theta}_i) \mathbf{1}(o_{cr}^{t_k}, \bar{o}_j)$

7) Probability for each output

$$\boxed{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i)} \leq p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i) \leq \boxed{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i)}$$

$$\frac{N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j}}{N_{\theta_{cr}=\bar{\theta}_i}} + Z_{\frac{r_\alpha}{2}} \sqrt{\frac{N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j} (N_{\theta_{cr}=\bar{\theta}_i} - N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j})}{N_{\theta_{cr}=\bar{\theta}_i}^3}} \quad \frac{N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j}}{N_{\theta_{cr}=\bar{\theta}_i}} - Z_{\frac{r_\alpha}{2}} \sqrt{\frac{N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j} (N_{\theta_{cr}=\bar{\theta}_i} - N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j})}{N_{\theta_{cr}=\bar{\theta}_i}^3}}$$

explain 7: confidence level r_α including in user parameter.

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

8) Notation

- prior knowledge about the probability of env(cr): $\sum_i^{N_e} r_{ep,k} = 1$

9) Posterior probability (Bayes rule)

$$p(\theta_{cr} = \bar{\theta}_i | o_{cr} = \bar{o}_j) = \frac{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i) p(\theta_{cr} = \bar{\theta}_i)}{p(o_{cr} = \bar{o}_j)} = \frac{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i) r_{ep,i}}{\sum_{k=1}^{N_e} p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_k) r_{ep,k}}$$

$$\overline{p(\theta_{cr} = \bar{\theta}_i | o_{cr} = \bar{o}_j)} := \frac{\overline{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i) r_{ep,i}}}{\sum_{k=1}^{N_e} \overline{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_k) r_{ep,k}}}$$

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

8) Notation

- prior knowledge about the probability of env(cr): $\sum_i^{N_e} \boxed{r_{ep,k}} = 1$

explain 8: also include in user parameter
→ *context*

9) Posterior probability (Bayes rule)

$$p(\theta_{cr} = \bar{\theta}_i | o_{cr} = \bar{o}_j) = \frac{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i) p(\theta_{cr} = \bar{\theta}_i)}{p(o_{cr} = \bar{o}_j)} = \frac{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i) r_{ep,i}}{\sum_{k=1}^{N_e} p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_k) r_{ep,k}}$$

$$\overline{p(\theta_{cr} = \bar{\theta}_i | o_{cr} = \bar{o}_j)} := \frac{\overline{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i) r_{ep,i}}}{\sum_{k=1}^{N_e} \overline{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_k) r_{ep,k}}}$$

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

10) Chance-constraint (replacement)

$$\forall i, Pr(c_i(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}_{cr}, \mathbf{r}) < 0) \leq r_{t,i}$$

$$\Rightarrow \sum_{k: c(\mathbf{x}, \mathbf{z}; \bar{\boldsymbol{\theta}}_k, \mathbf{r}) < 0} \overline{p(\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_k | \mathbf{o}_{cr} = \mathbf{o}_{cr}^r)} \leq r_{t,i}$$

$$\Rightarrow \begin{aligned} c(\mathbf{x}, \mathbf{z}; \bar{\boldsymbol{\theta}}_k, \mathbf{r}) + Ms_k &\geq 0, \quad k = 1, \dots, N_e \\ \sum_k \overline{p(\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_k | \mathbf{o}_{cr} = \mathbf{o}_{cr}^r)} s_k &\leq r_{t,i} \end{aligned}$$

$$\bar{c}_i(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) := \max_{s_1, \dots, s_{N_e} \in \{0,1\}} \min_k c(\mathbf{x}, \mathbf{z}; \bar{\boldsymbol{\theta}}_k, \mathbf{r}) + Ms_k$$
$$\sum_k \overline{p(\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_k | \mathbf{o}_{cr} = \mathbf{o}_{cr}^r)} s_k \leq r_{t,i}$$

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

10) Chance-constraint (replacement)

remind 1: environment-dependent constraint c_i for $i = n_{cg} + 1, \dots, n_c$

$$\boxed{\forall i, Pr(c_i(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}_{cr}, \mathbf{r}) < 0) \leq r_{t,i}}$$

$$\Rightarrow \sum_{k: c(\mathbf{x}, \mathbf{z}; \bar{\boldsymbol{\theta}}_k, \mathbf{r}) < 0} \overline{p(\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_k | \mathbf{o}_{cr} = \mathbf{o}_{cr}^r)} \leq r_{t,i}$$

$$\Rightarrow \begin{aligned} c(\mathbf{x}, \mathbf{z}; \bar{\boldsymbol{\theta}}_k, \mathbf{r}) + Ms_k &\geq 0, \quad k = 1, \dots, N_e \\ \sum_k \overline{p(\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_k | \mathbf{o}_{cr} = \mathbf{o}_{cr}^r)} s_k &\leq r_{t,i} \end{aligned}$$

$$\bar{c}_i(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) := \max_{s_1, \dots, s_{N_e} \in \{0,1\}} \min_k c(\mathbf{x}, \mathbf{z}; \bar{\boldsymbol{\theta}}_k, \mathbf{r}) + Ms_k$$
$$\sum_k \overline{p(\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_k | \mathbf{o}_{cr} = \mathbf{o}_{cr}^r)} s_k \leq r_{t,i}$$

Section 3: Dealing Constraints: Chance-Constrained Method

2. Chance-constrained method

10) Chance-constraint (replacement)

$$\forall i, Pr(c_i(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}_{cr}, \mathbf{r}) < 0) \leq r_{t,i}$$



$$\sum_{k: c(\mathbf{x}, \mathbf{z}; \bar{\boldsymbol{\theta}}_k, \mathbf{r}) < 0} \overline{p(\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_k | \mathbf{o}_{cr} = \mathbf{o}_{cr}^r)} \leq r_{t,i}$$

explain 9: sufficiently big constant.



$$c(\mathbf{x}, \mathbf{z}; \bar{\boldsymbol{\theta}}_k, \mathbf{r}) + \boxed{M}s_k \geq 0, \quad k = 1, \dots, N_e$$
$$\sum_k \overline{p(\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_k | \mathbf{o}_{cr} = \mathbf{o}_{cr}^r)} \boxed{s_k} \leq r_{t,i}$$

explain 10: integer variable to identify neglecting env.

$$\bar{c}_i(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) := \max_{s_1, \dots, s_{N_e} \in \{0,1\}} \min_k c(\mathbf{x}, \mathbf{z}; \bar{\boldsymbol{\theta}}_k, \mathbf{r}) + Ms_k$$
$$\sum_k \overline{p(\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_k | \mathbf{o}_{cr} = \mathbf{o}_{cr}^r)} s_k \leq r_{t,i}$$

Section 4: Training with Internal Test Cases

Section 4: Training with Internal Test Cases

1. Defining real performance

Condition 1

- a) sample space $\mathcal{Y}_{cr}$: The set of possible y_{cr}
- b) sigma-algebra $\Sigma_{\mathcal{Y}_{cr}}$
- c) probability measure $P_{\theta_{cr}} : \Sigma_{\mathcal{Y}_{cr}} \rightarrow [0, 1]$
- d) with Metric $M_{\mathcal{Y}_{cr}}$, $M : \Omega \times \Omega \rightarrow [0, \infty]$
- e) the postprocessing system (NN) classifies the data into finite outputs:

finite partition of $\mathcal{Y}_{cr}$: $Y_{\bar{o}_1}, \dots, Y_{\bar{o}_{N_o}}$ and $Y_{\bar{o}_j} \in \Sigma_{\mathcal{Y}_{cr}}$.

Section 4: Training with Internal Test Cases

1. Defining real performance

Condition 1

a) sample space $\mathcal{Y}_{cr}$: The set of possible y_{cr}

b) sigma-algebra $\Sigma_{\mathcal{Y}_{cr}}$

c) probability measure $P_{\theta_{cr}} : \Sigma_{\mathcal{Y}_{cr}} \rightarrow [0, 1]$

d) with Metric $M_{\mathcal{Y}_{cr}}$, $M : \Omega \times \Omega \rightarrow [0, \infty]$

e) the postprocessing system (NN) classifies the data into finite outputs:

finite partition of $\mathcal{Y}_{cr}$: $Y_{\bar{o}_1}, \dots, Y_{\bar{o}_{N_o}}$ and $Y_{\bar{o}_j} \in \Sigma_{\mathcal{Y}_{cr}}$.

Probability space

A probability space W is a unique triple $W = \{\Omega, \mathcal{F}, P\}$:

- Ω is its sample space
- $\mathcal{F}$ its σ -algebra of events
- P its probability measure

explain 1: to build probability space.

Section 4: Training with Internal Test Cases

1. Defining real performance

Condition 1

- a) sample space $\mathcal{Y}_{cr}$: The set of possible y_{cr}
- b) sigma-algebra $\Sigma_{\mathcal{Y}_{cr}}$
- c) probability measure $P_{\theta_{cr}} : \Sigma_{\mathcal{Y}_{cr}} \rightarrow [0, 1]$
- d) with Metric $M_{\mathcal{Y}_{cr}}$, $M : \Omega \times \Omega \rightarrow [0, \infty]$ **explain 2: to define open balls.**
- e) the postprocessing system (NN) classifies the data into finite outputs:

finite partition of $\mathcal{Y}_{cr}$: $Y_{\bar{o}_1}, \dots, Y_{\bar{o}_{N_o}}$ and $Y_{\bar{o}_j} \in \Sigma_{\mathcal{Y}_{cr}}$.

Section 4: Training with Internal Test Cases

1. Defining real performance

Condition 1

- a) sample space $\mathcal{Y}_{cr}$: The set of possible y_{cr}
- b) sigma-algebra $\Sigma_{\mathcal{Y}_{cr}}$
- c) probability measure $P_{\theta_{cr}} : \Sigma_{\mathcal{Y}_{cr}} \rightarrow [0, 1]$
- d) with Metric $M_{\mathcal{Y}_{cr}}$, $M : \Omega \times \Omega \rightarrow [0, \infty]$
- e) the postprocessing system (NN) classifies the data into finite outputs:

finite partition of $\mathcal{Y}_{cr}$: $Y_{\bar{o}_1}, \dots, Y_{\bar{o}_{N_o}}$ and $Y_{\bar{o}_j} \in \Sigma_{\mathcal{Y}_{cr}}$.

explain 3: map to the finite $\mathcal{O}_{cr}$

→ The real performance of the perception module: $p^(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i) := P_{\bar{\theta}_i}(Y_{\bar{o}_j})$*

Section 4: Training with Internal Test Cases

1. Defining real performance

1) Replacement of constraints:

$$\begin{aligned} \overline{p(\theta_{cr} = \bar{\theta}_i | o_{cr} = \bar{o}_j)} &:= \frac{\overline{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i) r_{ep,i}}}{\sum_{k=1}^{N_e} \overline{p(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_k) r_{ep,k}}} & \rightarrow & p^*(\theta_{cr} = \bar{\theta}_i | o_{cr} = \bar{o}_j) := \frac{p^*(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_i) r_{ep,i}}{\sum_{k=1}^{N_e} p^*(o_{cr} = \bar{o}_j | \theta_{cr} = \bar{\theta}_k) r_{ep,k}} \\ \bar{c}_i(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) &:= \max_{\substack{s_1, \dots, s_{N_e} \in \{0,1\}}} \min_k c(\mathbf{x}, \mathbf{z}; \bar{\theta}_k, \mathbf{r}) + M s_k \\ &\quad \sum_k \overline{p(\theta_{cr} = \bar{\theta}_k | o_{cr} = \bar{o}_{cr}^r) s_k} \leq r_{t,i} & \rightarrow & \bar{c}_i^*(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) := \max_{\substack{s_1, \dots, s_{N_e} \in \{0,1\}}} \min_k c(\mathbf{x}, \mathbf{z}; \bar{\theta}_k, \mathbf{r}) + M s_k \\ &\quad \sum_k p^*(\theta_{cr} = \bar{\theta}_k | o_{cr} = \bar{o}_{cr}^r) s_k \leq r_{t,i} \end{aligned}$$

Section 4: Training with Internal Test Cases

1. Defining real performance

2) We want to know condition of the ‘good’ internal test.

Definition 1

- a) compact metric space: Ω with $M : \Omega \times \Omega \rightarrow [0, \infty]$
- b) sigma-algebra that includes all open balls: Σ
- c) probability measure $P : \Sigma \rightarrow [0, 1]$
- d) we call a finite sequence of elements of Ω , $(\omega_1, \dots, \omega_{n_s})$, is ζ -informative if and only if

$$\text{for all subset } X \text{ of } \{1, \dots, n_s\}, \quad P(\cup_{i \in X} B(\omega_i, \zeta)) \geq \frac{|X|}{n_s}$$

Section 4: Training with Internal Test Cases

1. Defining real performance

2) Define the condition of ‘good’ internal test

Definition 1

a) compact metric space: Ω with $M : \Omega \times \Omega \rightarrow [0, \infty]$

b) sigma-algebra that includes all open balls: Σ

c) probability measure $P : \Sigma \rightarrow [0, 1]$

d) we call a finite sequence of elements of Ω , $(\omega_1, \dots, \omega_{n_s})$, is ζ -informative if and only if

$$\text{for all subset } X \text{ of } \{1, \dots, n_s\}, \quad P(\cup_{i \in X} B(\omega_i, \zeta)) \geq \frac{|X|}{n_s}$$

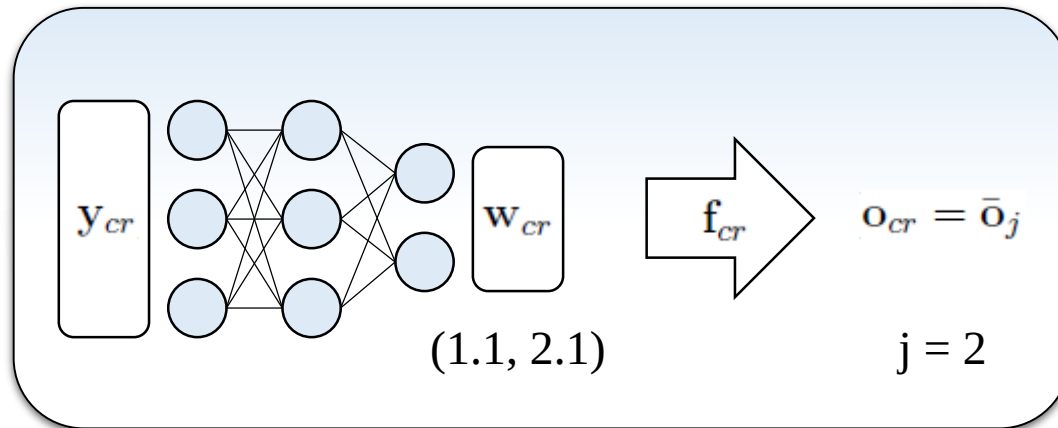
explain 4: margin ζ is robust to the selection of the subset of data.

this definition provides robustness for any network or parameters.

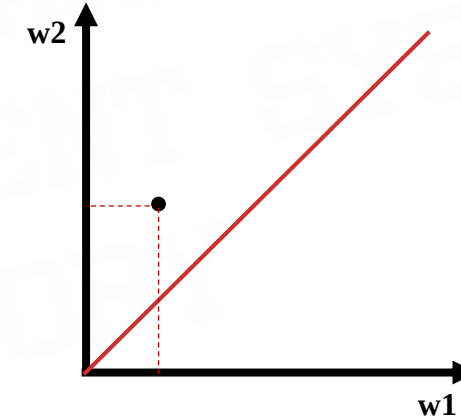
Section 4: Training with Internal Test Cases

2. Conservative response

1) Simple classification example



$$1(o_{cr}^{t_k}, \bar{o}_j) = 1(f_{cr}(w_{cr}^{t_k}), \bar{o}_j)$$



Section 4: Training with Internal Test Cases

2. Conservative response

2) Notation (remind + new)

- original: $1(o_{cr}^{t_k}, \bar{o}_j) = 1(f_{cr}(w_{cr}^{t_k}), \bar{o}_j)$
- ‘+xi term’: $1^{+\xi}(o_{cr}^{t_k}, \bar{o}_j) = \max_{\|\tilde{w}_{cr} - w_{cr}^{t_k}\| \leq \xi} 1(f_{cr}(\tilde{w}_{cr}), \bar{o}_j)$
- ‘-xi term’: $1^{-\xi}(o_{cr}^{t_k}, \bar{o}_j) = \min_{\|\tilde{w}_{cr} - w_{cr}^{t_k}\| \leq \xi} 1(f_{cr}(\tilde{w}_{cr}), \bar{o}_j)$

Section 4: Training with Internal Test Cases

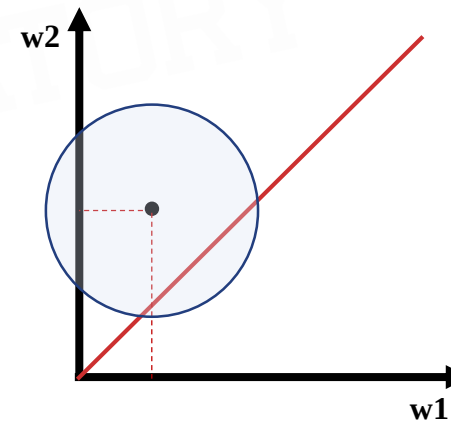
2. Conservative response

2) Notation (remind + new)

- original: $1(o_{cr}^{t_k}, \bar{o}_j) = 1(f_{cr}(w_{cr}^{t_k}), \bar{o}_j)$
- '+xi term': $1^{+\xi}(o_{cr}^{t_k}, \bar{o}_j) = \max_{\|\tilde{w}_{cr} - w_{cr}^{t_k}\| \leq \xi} 1(f_{cr}(\tilde{w}_{cr}), \bar{o}_j)$
- '-xi term': $1^{-\xi}(o_{cr}^{t_k}, \bar{o}_j) = \min_{\|\tilde{w}_{cr} - w_{cr}^{t_k}\| \leq \xi} 1(f_{cr}(\tilde{w}_{cr}), \bar{o}_j)$

example 1: big xi

- original: 1 at (j = 2)
- '+xi term': 1 at (j = 1, 2)
- '-xi term': 0



Section 4: Training with Internal Test Cases

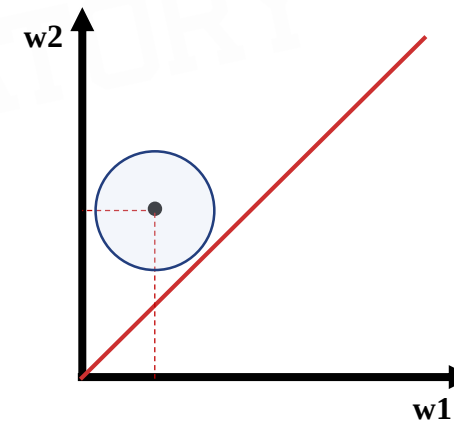
2. Conservative response

2) Notation (remind + new)

- original: $1(o_{cr}^{t_k}, \bar{o}_j) = 1(f_{cr}(w_{cr}^{t_k}), \bar{o}_j)$
- '+xi term': $1^{+\xi}(o_{cr}^{t_k}, \bar{o}_j) = \max_{\|\tilde{w}_{cr} - w_{cr}^{t_k}\| \leq \xi} 1(f_{cr}(\tilde{w}_{cr}), \bar{o}_j)$
- '-xi term': $1^{-\xi}(o_{cr}^{t_k}, \bar{o}_j) = \min_{\|\tilde{w}_{cr} - w_{cr}^{t_k}\| \leq \xi} 1(f_{cr}(\tilde{w}_{cr}), \bar{o}_j)$

example 2: small xi

- original: 1 at (j = 2)
- '+xi term': 1 at (j = 2)
- '-xi term': 1 at (j = 2)



Section 4: Training with Internal Test Cases

2. Conservative response

2) Notation (remind + new)

- original: $1(o_{cr}^{t_k}, \bar{o}_j) = 1(f_{cr}(w_{cr}^{t_k}), \bar{o}_j)$
- '+xi term': $1^{+\xi}(o_{cr}^{t_k}, \bar{o}_j) = \max_{\|\tilde{w}_{cr} - w_{cr}^{t_k}\| \leq \xi} 1(f_{cr}(\tilde{w}_{cr}), \bar{o}_j)$
- '-xi term': $1^{-\xi}(o_{cr}^{t_k}, \bar{o}_j) = \min_{\|\tilde{w}_{cr} - w_{cr}^{t_k}\| \leq \xi} 1(f_{cr}(\tilde{w}_{cr}), \bar{o}_j)$

explain 1: $1^{-\xi}(o_{cr}^{t_k}, \bar{o}_j) \leq 1(o_{cr}^{t_k}, \bar{o}_j) \leq 1^{+\xi}(o_{cr}^{t_k}, \bar{o}_j)$

Section 4: Training with Internal Test Cases

2. Conservative response

3) Notation (remind + new)

- number of internal test cases for env and output: $N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j} = \sum_{k=1}^{N_t} \mathbf{1}(\theta_{cr}^{t_k}, \bar{\theta}_i) \mathbf{1}(o_{cr}^{t_k}, \bar{o}_j)$
- new 1: $N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j}^{+\xi} = \sum_{k=1}^{N_t} \mathbf{1}(\theta_{cr}^{t_k}, \bar{\theta}_i) \mathbf{1}^{+\xi}(o_{cr}^{t_k}, \bar{o}_j)$
- new 2: $N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j}^{-\xi} = \sum_{k=1}^{N_t} \mathbf{1}(\theta_{cr}^{t_k}, \bar{\theta}_i) \mathbf{1}^{-\xi}(o_{cr}^{t_k}, \bar{o}_j)$

Section 4: Training with Internal Test Cases

2. Conservative response

3) Notation (remind + new)

- number of internal test cases for env and output: $N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j} = \sum_{k=1}^{N_t} \mathbf{1}(\theta_{cr}^{t_k}, \bar{\theta}_i) \mathbf{1}(o_{cr}^{t_k}, \bar{o}_j)$

- new 1: $N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j}^{+\xi} = \sum_{k=1}^{N_t} \mathbf{1}(\theta_{cr}^{t_k}, \bar{\theta}_i) \mathbf{1}^{+\xi}(o_{cr}^{t_k}, \bar{o}_j)$

- new 2: $N_{\theta_{cr}=\bar{\theta}_i, o_{cr}=\bar{o}_j}^{-\xi} = \sum_{k=1}^{N_t} \mathbf{1}(\theta_{cr}^{t_k}, \bar{\theta}_i) \mathbf{1}^{-\xi}(o_{cr}^{t_k}, \bar{o}_j)$

explain 2: upper bound and lower bound

Section 4: Training with Internal Test Cases

2. Conservative response

4) Notation

$$p^{+\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) := \frac{N_{\boldsymbol{\theta}_{cr}=\bar{\boldsymbol{\theta}}_i, \mathbf{o}_{cr}=\bar{\mathbf{o}}_j}^{+\xi}}{N_{\boldsymbol{\theta}_{cr}=\bar{\boldsymbol{\theta}}_i}}, p^{-\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) := \frac{N_{\boldsymbol{\theta}_{cr}=\bar{\boldsymbol{\theta}}_i, \mathbf{o}_{cr}=\bar{\mathbf{o}}_j}^{-\xi}}{N_{\boldsymbol{\theta}_{cr}=\bar{\boldsymbol{\theta}}_i}}$$

$$p^{\xi}(\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i | \mathbf{o}_{cr} = \bar{\mathbf{o}}_j) := \frac{p^{+\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) r_{ep,i}}{\sum_{k=1}^{N_e} p^{-\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_k) r_{ep,k}}$$

Section 4: Training with Internal Test Cases

2. Conservative response

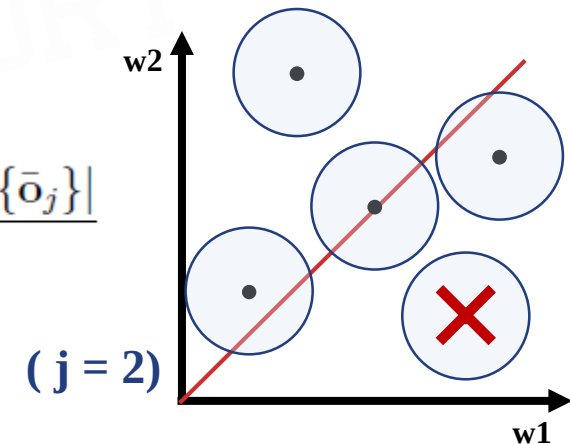
4) Notation

$$p^{+\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) := \frac{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i, \mathbf{o}_{cr} = \bar{\mathbf{o}}_j}^{+\xi}}{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i}}, p^{-\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) := \frac{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i, \mathbf{o}_{cr} = \bar{\mathbf{o}}_j}^{-\xi}}{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i}}$$

$$p^{\xi}(\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i | \mathbf{o}_{cr} = \bar{\mathbf{o}}_j) := \frac{p^{+\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) r_{ep,i}}{\sum_{k=1}^{N_e} p^{-\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_k) r_{ep,k}}$$

explain 3: by definition,

$$p^{+\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) = \frac{|\{k | \boldsymbol{\theta}_{cr}^{t_k} = \bar{\boldsymbol{\theta}}_i, \mathbf{f}_{cr}(\bar{B}(\mathbf{w}_{cr}(\mathbf{y}_{cr}^{t_k}), \xi)) \supset \{\bar{\mathbf{o}}_j\}\}|}{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i}}$$



Section 4: Training with Internal Test Cases

2. Conservative response

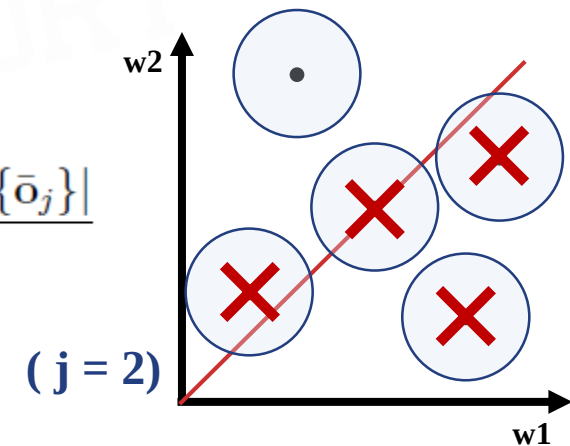
4) Notation

$$p^{+\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) := \frac{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i, \mathbf{o}_{cr} = \bar{\mathbf{o}}_j}^{+\xi}}{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i}}, \quad p^{-\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) := \frac{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i, \mathbf{o}_{cr} = \bar{\mathbf{o}}_j}^{-\xi}}{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i}}$$

$$p^{\xi}(\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i | \mathbf{o}_{cr} = \bar{\mathbf{o}}_j) := \frac{p^{+\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) r_{ep,i}}{\sum_{k=1}^{N_e} p^{-\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_k) r_{ep,k}}$$

explain 4: by definition,

$$p^{-\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) = \frac{|\{k | \boldsymbol{\theta}_{cr}^{t_k} = \bar{\boldsymbol{\theta}}_i, \mathbf{f}_{cr}(\bar{B}(\mathbf{w}_{cr}(\mathbf{y}_{cr}^{t_k}), \xi)) = \{\bar{\mathbf{o}}_j\}\}|}{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i}}$$



Section 4: Training with Internal Test Cases

2. Conservative response

5) Conservative replacement of constraints

$$\bar{c}_i^\xi(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) := \max_{\substack{s_1, \dots, s_{N_e} \in \{0,1\} \\ \sum_k p^\xi(\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_k | \mathbf{o}_{cr} = \mathbf{o}_{cr}^r) s_k \leq r_{t,i}}} \min_k c_i(\mathbf{x}, \mathbf{z}; \bar{\boldsymbol{\theta}}_k, \mathbf{r}) + M s_k$$

Section 4: Training with Internal Test Cases

2. Conservative response

6) We can add the new condition to guarantees the conservativeness

Condition 2 For some (ζ, ξ) ,

- a) $\Sigma_{\mathcal{Y}_{cr}}$ include all open balls based on $M_{\mathcal{Y}_{cr}}$.
- b) For each $\bar{\theta}_i$, subsequence of $(\mathbf{y}_{cr}^{t_1}, \dots, \mathbf{y}_{cr}^{t_{N_t}})$, that is, $(\mathbf{y}_{cr}^{t_{a_1}}, \dots, \mathbf{y}_{cr}^{t_{a_{N_{at}}}}) : \{a_j\} = \{k | \theta_{cr}^{t_k} = \bar{\theta}_i\}$,
is ζ -informative in $(\mathcal{Y}_{cr}, M_{\mathcal{Y}_{cr}}, \Sigma_{\mathcal{Y}_{cr}}, P_{\bar{\theta}_i})$.
- c) $\mathbf{w}_{cr}(\mathbf{y}_{cr})$ satisfies that $\|\mathbf{w}_{cr}(v_1) - \mathbf{w}_{cr}(v_2)\| \leq \xi$ if $M_{\mathcal{Y}_{cr}}(v_1, v_2) < \zeta$ for any $v_1, v_2 \in \mathcal{Y}_{cr}$.

Section 4: Training with Internal Test Cases

2. Conservative response

6) We can add the new condition to guarantees the conservativeness

Condition 2 For some (ζ, ξ) ,

- a) $\Sigma_{\mathcal{Y}_{cr}}$ include all open balls based on $M_{\mathcal{Y}_{cr}}$. **explain 5: to follow Definition 1.**
- b) For each $\bar{\theta}_i$, subsequence of $(\mathbf{y}_{cr}^{t_1}, \dots, \mathbf{y}_{cr}^{t_{N_t}})$, that is, $(\mathbf{y}_{cr}^{t_{a_1}}, \dots, \mathbf{y}_{cr}^{t_{a_{N_{at}}}}) : \{a_j\} = \{k | \theta_{cr}^{t_k} = \bar{\theta}_i\}$, is ζ -informative in $(\mathcal{Y}_{cr}, M_{\mathcal{Y}_{cr}}, \Sigma_{\mathcal{Y}_{cr}}, P_{\bar{\theta}_i})$.
- c) $\mathbf{w}_{cr}(\mathbf{y}_{cr})$ satisfies that $\|\mathbf{w}_{cr}(v_1) - \mathbf{w}_{cr}(v_2)\| \leq \xi$ if $M_{\mathcal{Y}_{cr}}(v_1, v_2) < \zeta$ for any $v_1, v_2 \in \mathcal{Y}_{cr}$.

Section 4: Training with Internal Test Cases

2. Conservative response

6) We can add the new condition to guarantees the conservativeness

Condition 2 For some (ζ, ξ) ,

a) $\Sigma_{\mathcal{Y}_{cr}}$ include all open balls based on $M_{\mathcal{Y}_{cr}}$.

b) For each $\bar{\theta}_i$, subsequence of $(\mathbf{y}_{cr}^{t_1}, \dots, \mathbf{y}_{cr}^{t_{N_t}})$, that is, $(\mathbf{y}_{cr}^{t_{a_1}}, \dots, \mathbf{y}_{cr}^{t_{a_{N_{at}}}}) : \{a_j\} = \{k | \theta_{cr}^{t_k} = \bar{\theta}_i\}$, is ζ -informative in $(\mathcal{Y}_{cr}, M_{\mathcal{Y}_{cr}}, \Sigma_{\mathcal{Y}_{cr}}, P_{\bar{\theta}_i})$.

c) $\mathbf{w}_{cr}(\mathbf{y}_{cr})$ satisfies that $\|\mathbf{w}_{cr}(v_1) - \mathbf{w}_{cr}(v_2)\| \leq \xi$ if $M_{\mathcal{Y}_{cr}}(v_1, v_2) < \zeta$ for any $v_1, v_2 \in \mathcal{Y}_{cr}$.

explain 6:

- big ζ : many sequences are ζ -informative.
- small ζ : only high-quality sequences (internal test cases) are ζ -informative.

Section 4: Training with Internal Test Cases

2. Conservative response

6) We can add the new condition to guarantees the conservativeness

Condition 2 For some (ζ, ξ) ,

- a) $\Sigma_{\mathcal{Y}_{cr}}$ include all open balls based on $M_{\mathcal{Y}_{cr}}$.
- b) For each $\bar{\theta}_i$, subsequence of $(y_{cr}^{t_1}, \dots, y_{cr}^{t_{N_t}})$, that is, $(y_{cr}^{t_{a_1}}, \dots, y_{cr}^{t_{a_{N_{at}}}}) : \{a_j\} = \{k | \theta_{cr}^{t_k} = \bar{\theta}_i\}$, is ζ -informative in $(\mathcal{Y}_{cr}, M_{\mathcal{Y}_{cr}}, \Sigma_{\mathcal{Y}_{cr}}, P_{\bar{\theta}_i})$.
- c) $w_{cr}(y_{cr})$ satisfies that $\|w_{cr}(v_1) - w_{cr}(v_2)\| \leq \xi$ if $M_{\mathcal{Y}_{cr}}(v_1, v_2) < \zeta$ for any $v_1, v_2 \in \mathcal{Y}_{cr}$.

explain 6, 7:

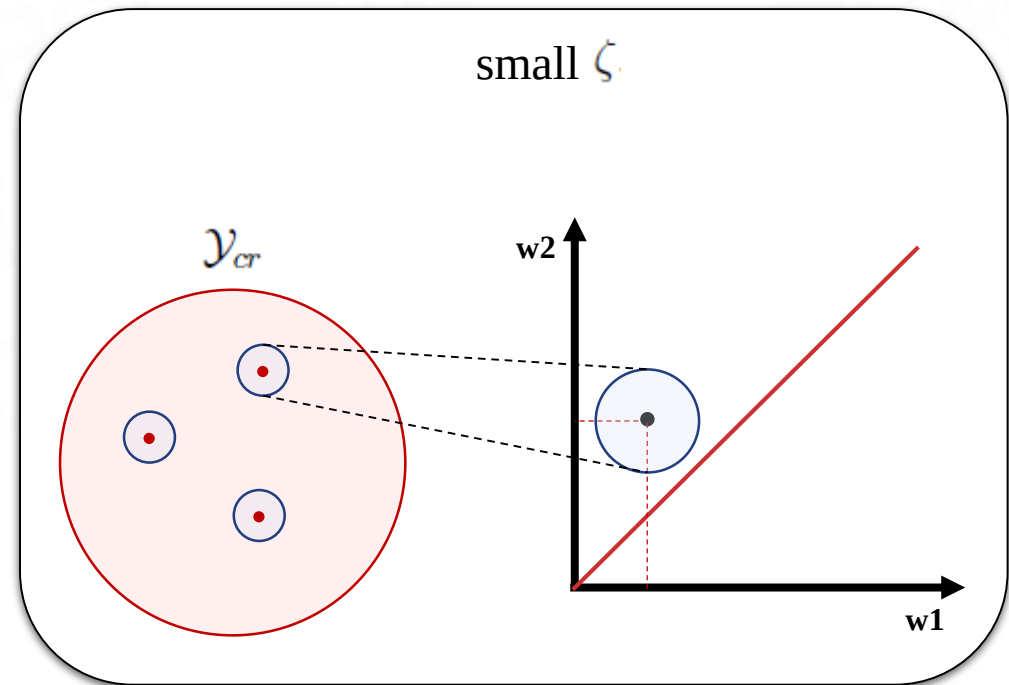
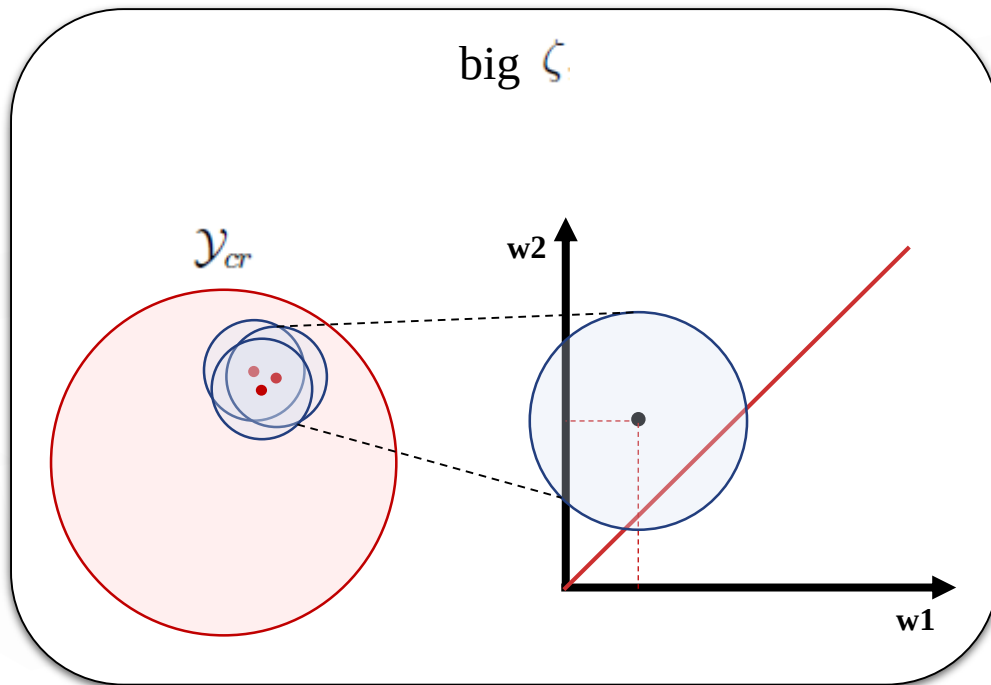
- big ζ : many sequences are ζ -informative. However, big ξ makes much conservative control.
- small ζ : only high-quality sequences (internal test cases) are ζ -informative.

However, small ξ makes less conservative control.

Section 4: Training with Internal Test Cases

2. Conservative response

7) Simple intuition: both of them are zeta-informative.



Section 4: Training with Internal Test Cases

2. Conservative response

Theorem 1 (with Condition 1 & 2)

$$p^{-\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) \leq p^*(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) \leq p^{+\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i)$$

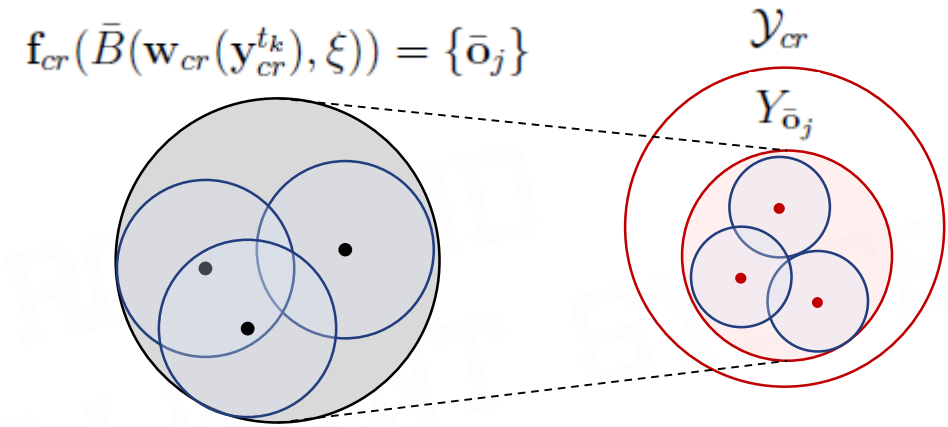
A part of Proof (first inequality):

$$\begin{aligned} p^{-\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) &= \frac{|\{k | \boldsymbol{\theta}_{cr}^{t_k} = \bar{\boldsymbol{\theta}}_i, f(\bar{B}(\mathbf{w}_{cr}(\mathbf{y}_{cr}^{t_k}), \xi)) = \{\bar{\mathbf{o}}_j\}\}|}{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i}} \leq \frac{|\{k | \boldsymbol{\theta}_{cr}^{t_k} = \bar{\boldsymbol{\theta}}_i, B(\mathbf{y}_{cr}^{t_k}, \zeta) \subset Y_{\bar{\mathbf{o}}_j}\}|}{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i}} \\ &\leq P_{\bar{\boldsymbol{\theta}}_i}(\cup_{\boldsymbol{\theta}_{cr}^{t_k} = \bar{\boldsymbol{\theta}}_i, B(\mathbf{y}_{cr}^{t_k}, \zeta) \subset Y_{\bar{\mathbf{o}}_j}} B(\mathbf{y}_{cr}^{t_k}, \zeta)) \leq P_{\bar{\boldsymbol{\theta}}_i}(Y_{\bar{\mathbf{o}}_j}) = p^*(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) \end{aligned}$$

Section 4: Training with Internal Test Cases

2. Conservative response

explain 8:



A part of Proof (first inequality):

by Condition 2-c

$$p^{-\xi}(\mathbf{o}_{cr} = \bar{o}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) = \frac{|\{k | \boldsymbol{\theta}_{cr}^{t_k} = \bar{\boldsymbol{\theta}}_i, f(\bar{B}(\mathbf{w}_{cr}(\mathbf{y}_{cr}^{t_k}), \xi)) = \{\bar{o}_j\}\}|}{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i}} \boxed{\leq} \frac{|\{k | \boldsymbol{\theta}_{cr}^{t_k} = \bar{\boldsymbol{\theta}}_i, B(\mathbf{y}_{cr}^{t_k}, \zeta) \subset Y_{\bar{o}_j}\}|}{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i}}$$

$$\leq P_{\bar{\boldsymbol{\theta}}_i}(\cup_{\boldsymbol{\theta}_{cr}^{t_k} = \bar{\boldsymbol{\theta}}_i, B(\mathbf{y}_{cr}^{t_k}, \zeta) \subset Y_{\bar{o}_j}} B(\mathbf{y}_{cr}^{t_k}, \zeta)) \leq P_{\bar{\boldsymbol{\theta}}_i}(Y_{\bar{o}_j}) = p^*(\mathbf{o}_{cr} = \bar{o}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i)$$

Section 4: Training with Internal Test Cases

2. Conservative response

Theorem 1 (with Condition 1 & 2)

$$p^{-\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) \leq p^*(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) \leq p^{+\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i)$$

A part of Proof (first inequality):

$$p^{-\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) = \frac{|\{k | \boldsymbol{\theta}_{cr}^{t_k} = \bar{\boldsymbol{\theta}}_i, f(\bar{B}(\mathbf{w}_{cr}(\mathbf{y}_{cr}^{t_k}), \xi)) = \{\bar{\mathbf{o}}_j\}\}|}{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i}} \leq \frac{|\{k | \boldsymbol{\theta}_{cr}^{t_k} = \bar{\boldsymbol{\theta}}_i, B(\mathbf{y}_{cr}^{t_k}, \zeta) \subset Y_{\bar{\mathbf{o}}_j}\}|}{N_{\boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i}}$$

$$\boxed{\leq} P_{\bar{\boldsymbol{\theta}}_i}(\cup_{\boldsymbol{\theta}_{cr}^{t_k} = \bar{\boldsymbol{\theta}}_i, B(\mathbf{y}_{cr}^{t_k}, \zeta) \subset Y_{\bar{\mathbf{o}}_j}} B(\mathbf{y}_{cr}^{t_k}, \zeta)) \leq P_{\bar{\boldsymbol{\theta}}_i}(Y_{\bar{\mathbf{o}}_j}) = p^*(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i)$$

by Condition 2-b

explain 9:

Section 4: Training with Internal Test Cases

2. Conservative response

Theorem 1 (with Condition 1 & 2)

$$p^{-\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) \leq p^*(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i) \leq p^{+\xi}(\mathbf{o}_{cr} = \bar{\mathbf{o}}_j | \boldsymbol{\theta}_{cr} = \bar{\boldsymbol{\theta}}_i)$$

8) Corollary 1 is straightforward from Theorem 1.

Corollary 1 (with Condition 1 & 2)

$$\bar{c}_i^{\xi}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \leq \bar{c}_i^*(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}).$$

Section 4: Training with Internal Test Cases

2. Conservative response

Definition 2

For three functions, $\pi, \chi, \psi : R^{n_{real}} \times Z^{n_{inte}} \rightarrow R$,

we call π **aligns well** with χ under constraint $\psi \geq 0$ in set $\mathcal{S} \subset R^{n_{real}} \times Z^{n_{inte}}$ when

for all $\alpha_1, \alpha_2 \in \mathcal{S}$, when $\chi(\alpha_1) \leq \chi(\alpha_2)$ and $\psi(\alpha_1) < 0 \leq \psi(\alpha_2)$, then $\pi(\alpha_1) \leq \pi(\alpha_2)$.

Condition 3

As functions of $(\mathbf{x}, \mathbf{z})$, for any c_i needed to be replaced by chance-constrained method and any $\bar{\theta}_k \in \Theta_{cr}$,

L **aligns well** with $\bar{J}$ under constraint $c_i \geq 0$ in $\mathcal{U}$.

Section 4: Training with Internal Test Cases

2. Conservative response

Definition 2

For three functions, $\pi, \chi, \psi : R^{n_{real}} \times Z^{n_{inte}} \rightarrow R$,

we call π **aligns well** with χ under constraint $\psi \geq 0$ in set $\mathcal{S} \subset R^{n_{real}} \times Z^{n_{inte}}$ when

for all $\alpha_1, \alpha_2 \in \mathcal{S}$, when $\chi(\alpha_1) \leq \chi(\alpha_2)$ and $\psi(\alpha_1) < 0 \leq \psi(\alpha_2)$, then $\pi(\alpha_1) \leq \pi(\alpha_2)$.

Condition 3

As functions of $(\mathbf{x}, \mathbf{z})$, for any c_i needed to be replaced by chance-constrained method and any $\bar{\theta}_k \in \Theta_{cr}$,

L **aligns well** with $\bar{J}$ under constraint $c_i \geq 0$ in $\mathcal{U}$.

explain 10:

- $\bar{J}$ is set to obtain high control performance and L is set to evaluate the performance.
- automatically satisfied when L is non-decreasing function of $\bar{J}$ or converse.

Section 4: Training with Internal Test Cases

2. Conservative response

Proposition 2 (with Condition 1 & 2 & 3)

The minimum of L with the optimal action is higher or equal when $\bar{c}_i^*$ are replaced with $\bar{c}_i^\xi$.

Section 4: Training with Internal Test Cases

2. Conservative response

Proposition 2 (with Condition 1 & 2 & 3)

The minimum of L with the optimal action is higher or equal when $\bar{c}_i^*$ are replaced with $\bar{c}_i^\xi$.

explain 11: ‘conservative response’

Section 4: Training with Internal Test Cases

2. Conservative response

Proposition 2 (with Condition 1 & 2 & 3)

The minimum of L with the optimal action is higher or equal when $\bar{c}_i^*$ are replaced with $\bar{c}_i^\xi$.

Proof :

- (1) with xi constraints $\bar{c}_i^\xi: (x_1^*, z_1^*)$
- (2) with (real) constraints $\bar{c}_i^*: (x_2^*, z_2^*)$
- (3) proposition 2 implies: $L(x_1^*, z_1^*) \geq L(x_2^*, z_2^*)$
- (4) assumption for contradiction: $L(x_1^*, z_1^*) < L(x_2^*, z_2^*)$

Section 4: Training with Internal Test Cases

2. Conservative response

Proposition 2 (with Condition 1 & 2 & 3)

The minimum of L with the optimal action is higher or equal when $\bar{c}_i^*$ are replaced with $\bar{c}_i^\xi$.

Proof: (4) assumption for contradiction: $L(\mathbf{x}_1^*, \mathbf{z}_1^*) < L(\mathbf{x}_2^*, \mathbf{z}_2^*)$

Theorem 1 & Corollary 1 = $\bar{J}(\mathbf{x}_1^*, \mathbf{z}_1^*) \geq \bar{J}(\mathbf{x}_2^*, \mathbf{z}_2^*)$ and ‘ $c_i(\mathbf{x}_2^*, \mathbf{z}_2^*) \geq 0 \rightarrow c_i(\mathbf{x}_1^*, \mathbf{z}_1^*) \geq 0$ ’.

Condition 3 (well-alignment) = $\bar{J}(\mathbf{x}_1^*, \mathbf{z}_1^*) \geq \bar{J}(\mathbf{x}_2^*, \mathbf{z}_2^*)$ and $c_i(\mathbf{x}_1^*, \mathbf{z}_1^*) \geq 0 > c_i(\mathbf{x}_2^*, \mathbf{z}_2^*)$, then $L(\mathbf{x}_1^*, \mathbf{z}_1^*) \geq L(\mathbf{x}_2^*, \mathbf{z}_2^*)$

Considering the assumption: $c_i(\mathbf{x}_1^*, \mathbf{z}_1^*) \geq 0 \rightarrow c_i(\mathbf{x}_2^*, \mathbf{z}_2^*) \geq 0$

Thus, $(\mathbf{x}_1^*, \mathbf{z}_1^*)$ and $(\mathbf{x}_2^*, \mathbf{z}_2^*)$ satisfy same set of original constraints and $\bar{J}(\mathbf{x}_1^*, \mathbf{z}_1^*) = \bar{J}(\mathbf{x}_2^*, \mathbf{z}_2^*)$.

Contradiction! $\therefore L(\mathbf{x}_1^*, \mathbf{z}_1^*) \geq L(\mathbf{x}_2^*, \mathbf{z}_2^*)$

Section 5: Construction of the Loss function for Training

Section 5: Construction of the Loss function for Training

1. General form of the loss function

1) Loss function

- Form: $L(\mathbf{x}, \mathbf{z}, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$
- In most cases, $J(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r})$ is used as loss function.
- Since loss function is the function of $\mathbf{x}$ and $\mathbf{z}$, it depends on the optimal solutions.

$$L(\mathbf{x}^*(\mathbf{o}, \mathbf{r}), \mathbf{z}^*(\mathbf{o}, \mathbf{r}), \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$$

Section 5: Construction of the Loss function for Training

1. General form of the loss function

1) Loss function

- Form: $L(\mathbf{x}, \mathbf{z}, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$

- In most cases, $J(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r})$ is used as loss function.
- Since loss function is the function of $\mathbf{x}$ and $\mathbf{z}$, it depends on the optimal solutions.

$$L(\mathbf{x}^*(\mathbf{o}, \mathbf{r}), \mathbf{z}^*(\mathbf{o}, \mathbf{r}), \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$$

explain 1: the loss function can be any function.

depend on user. ($\bar{J}$ or traditional loss functions)

Section 5: Construction of the Loss function for Training

1. General form of the loss function

1) Loss function

- Form: $L(\mathbf{x}, \mathbf{z}, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$
- In most cases, $J(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r})$ is used as loss function.
- Since loss function is the function of $\mathbf{x}$ and $\mathbf{z}$, it depends on the optimal solutions.

$$L(\mathbf{x}^*(\mathbf{o}, \mathbf{r}), \mathbf{z}^*(\mathbf{o}, \mathbf{r}), \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$$

issue 1: optimal solutions $\mathbf{x}^*(\mathbf{o}, \mathbf{r}), \mathbf{z}^*(\mathbf{o}, \mathbf{r})$ are not generally continuous functions of network output.

issue 2 - 1: multiple optimal solutions. corresponding loss can be different.

→ *Loss function cannot be well-defined.*

Section 5: Construction of the Loss function for Training

1. General form of the loss function

1) Loss function

- Form: $L(\mathbf{x}, \mathbf{z}, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$
- In most cases, $J(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}, \mathbf{r})$ is used as loss function.
- Since loss function is the function of $\mathbf{x}$ and $\mathbf{z}$, it depends on the optimal solutions.

$$L(\mathbf{x}^*(\mathbf{o}, \mathbf{r}), \mathbf{z}^*(\mathbf{o}, \mathbf{r}), \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$$

issue 2 - 2: how about well-defined loss function by choosing minimum value?

→ *Two stage minimization (computationally hard)*

→ *We cannot guarantee that we can have a closed-form solution.*

Section 5: Construction of the Loss function for Training

1. General form of the loss function

2) Alternative of the loss function

- approximated general loss function: $\tilde{L} \simeq L(\mathbf{x}^*(\mathbf{o}, \mathbf{r}), \mathbf{z}^*(\mathbf{o}, \mathbf{r}), \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$
- continuous with respect of the network output.
- well defined.
- avoid two-stage minimization (minimizing loss function within the optimal solution set)

Section 5: Construction of the Loss function for Training

2. Continuity issue

1) Continuity issue

- Main idea: optimal objective(approximated loss) is continuous w.r.t the parameters(network output).

a) objective function and constraints are continuous w.r.t the variables.

b) objective function is continuous w.r.t the parameters.

c) constraints are independent of the parameters.

Section 5: Construction of the Loss function for Training

2. Continuity issue

1) Continuity issue

- Main idea: optimal objective(approximated loss) is continuous w.r.t the parameters(network output).

a) objective function and constraints are continuous w.r.t the variables. *[Assumption 1, 2]*

b) objective function is continuous w.r.t the parameters. *[Assumption 2]*

c) constraints are independent of the parameters. *[Converted optimization problem]*

Section 5: Construction of the Loss function for Training

3. Continuity issue: first step

1) Converted optimization problem

Original optimization problem

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{X}, \mathbf{z} \in \mathcal{Z}} \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \\ \text{subject to } \bar{c}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \geq 0 \end{aligned}$$

Converted optimization problem

$$\min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \beta^\top \min(\bar{c}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), 0)$$

Section 5: Construction of the Loss function for Training

3. Continuity issue: first step

1) Converted optimization problem

Original optimization problem

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{X}, \mathbf{z} \in \mathcal{Z}} \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \\ \text{subject to } \bar{c}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \geq 0 \end{aligned}$$

Converted optimization problem

$$\min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \beta^\top \min(\bar{c}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), 0)$$

explain 1: constraints are independent of the parameters (Continuity issue - c)

Section 5: Construction of the Loss function for Training

3. Continuity issue: first step

1) Converted optimization problem

Original optimization problem

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{X}, \mathbf{z} \in \mathcal{Z}} \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \\ \text{subject to } \bar{c}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \geq 0 \end{aligned}$$

Converted optimization problem

$$\min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \beta^\top \min(\bar{c}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), 0)$$

explain 2: compact feasible region. prevent that problem is unbounded below.

Section 5: Construction of the Loss function for Training

3. Continuity issue: first step

1) Converted optimization problem

Original optimization problem

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{X}, \mathbf{z} \in \mathcal{Z}} \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \\ \text{subject to } \bar{c}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \geq 0 \end{aligned}$$

Converted optimization problem

$$\min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \beta^\top \min(\bar{c}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), 0)$$

issue 1: cannot guarantee the equivalence between the original and converted problem.

→ Check the distance between the solutions of these two problems. [Theorem 2]

Section 5: Construction of the Loss function for Training

3. Continuity issue: first step

2) Notation

- solution set of the original optimization problem: $S(\mathbf{o}, \mathbf{r}) := \underset{\mathbf{x}, \mathbf{z}}{\arg \min} \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$
s. t. $\bar{c}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \geq 0$
- solution space of specific $\mathbf{z}$: $S_{\mathbf{z}}(\mathbf{o}, \mathbf{r}) := \{\mathbf{x} : (\mathbf{x}, \mathbf{z}) \in S(\mathbf{o}, \mathbf{r})\}$
- $\tilde{S}_{\mathbf{z}}(\mathbf{o}, \mathbf{r}, \epsilon_1) = \bigcup_{\mathbf{x} \in S_{\mathbf{z}}(\mathbf{o}, \mathbf{r})} B(\mathbf{x}, \epsilon_1)$
- $\tilde{S}(\mathbf{o}, \mathbf{r}, \epsilon_1) = \bigcup_{\mathbf{z}} \tilde{S}_{\mathbf{z}}(\mathbf{o}, \mathbf{r}, \epsilon_1) \times \{\mathbf{z}\}$

Section 5: Construction of the Loss function for Training

3. Continuity issue: first step

2) Notation

- solution set of the original optimization problem: $S(\mathbf{o}, \mathbf{r}) := \underset{\mathbf{x}, \mathbf{z}}{\arg \min} \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$
s. t. $\bar{c}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \geq 0$
- solution space of specific $\mathbf{z}$: $S_{\mathbf{z}}(\mathbf{o}, \mathbf{r}) := \{\mathbf{x} : (\mathbf{x}, \mathbf{z}) \in S(\mathbf{o}, \mathbf{r})\}$
- $\tilde{S}_{\mathbf{z}}(\mathbf{o}, \mathbf{r}, \epsilon_1) = \bigcup_{\mathbf{x} \in S_{\mathbf{z}}(\mathbf{o}, \mathbf{r})} B(\mathbf{x}, \epsilon_1)$
- $\tilde{S}(\mathbf{o}, \mathbf{r}, \epsilon_1) = \bigcup_{\mathbf{z}} \tilde{S}_{\mathbf{z}}(\mathbf{o}, \mathbf{r}, \epsilon_1) \times \{\mathbf{z}\}$

explain 1: compact set.

Section 5: Construction of the Loss function for Training

3. Continuity issue: first step

Assumption 1

- a) $\bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{x}$.
- b) $\bar{c}_i(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{x}$.
- c) $\mathcal{U}$ is a finite union of multiplications of different elements of $\mathcal{Z}$ and paired compact subsets of $\mathcal{X}$.
$$\bigcup_1^{m_u(<\infty)} \mathcal{X}_i \times \{\mathbf{z}_i\}, \forall i, \mathcal{X}_i(\subset \mathcal{X}): \text{compact}, \forall i, j, \mathbf{z}_i \neq \mathbf{z}_j$$
- d) $\mathcal{U}$ includes all optimal solutions of the original optimization problem.

Section 5: Construction of the Loss function for Training

3. Continuity issue: first step

Assumption 1

- a) $\bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{x}$.
- b) $\bar{c}_i(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{x}$.
- c) $\mathcal{U}$ is a finite union of multiplications of different elements of $\mathcal{Z}$ and paired compact subsets of $\mathcal{X}$.
$$\bigcup_1^{m_u(<\infty)} \mathcal{X}_i \times \{\mathbf{z}_i\}, \forall i, \mathcal{X}_i(\subset \mathcal{X}): \text{compact}, \forall i, j, \mathbf{z}_i \neq \mathbf{z}_j$$
- d) $\mathcal{U}$ includes all optimal solutions of the original optimization problem.

explain 3: objective function and constraints are continuous w.r.t the variables. (Continuity issue - a)

Section 5: Construction of the Loss function for Training

3. Continuity issue: first step

Assumption 1

- a) $\bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{x}$.
- b) $\bar{c}_i(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{x}$.
- c) $\mathcal{U}$ is a finite union of multiplications of different elements of $\mathcal{Z}$ and paired compact subsets of $\mathcal{X}$.
$$\bigcup_1^{m_u(<\infty)} \mathcal{X}_i \times \{\mathbf{z}_i\}, \forall i, \mathcal{X}_i(\subset \mathcal{X}): \text{compact}, \forall i, j, \mathbf{z}_i \neq \mathbf{z}_j$$
- d) $\mathcal{U}$ includes all optimal solutions of the original optimization problem.

explain 4: finite unions of compact sets are compact.

Section 5: Construction of the Loss function for Training

3. Continuity issue: first step

Assumption 1

- a) $\bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{x}$.
- b) $\bar{c}_i(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{x}$.
- c) $\mathcal{U}$ is a finite union of multiplications of different elements of $\mathcal{Z}$ and paired compact subsets of $\mathcal{X}$.
$$\bigcup_1^{m_u(<\infty)} \mathcal{X}_i \times \{\mathbf{z}_i\}, \forall i, \mathcal{X}_i (\subset \mathcal{X}): \text{compact}, \forall i, j, \mathbf{z}_i \neq \mathbf{z}_j$$
- d) $\mathcal{U}$ includes all optimal solutions of the original optimization problem.

explain 5: Theorem 2 proof,

a) continuous + inverse image of closed set = closed set $H(\mathbf{o}, \mathbf{r}) = \{(\mathbf{x}, \mathbf{z}) \in \mathcal{U} | \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \leq \bar{J}^*(\mathbf{o}; \mathbf{r})\}$

b) intersection of a closed set with a compact set = compact set $(\mathcal{U}_z \cap H(\mathbf{o}, \mathbf{r})) \setminus \tilde{S}(\mathbf{o}, \mathbf{r}, \epsilon_1)$

c) continuous + compact domain = maximum value exist

$$\phi(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) : (\mathcal{U}_z \cap H(\mathbf{o}, \mathbf{r})) \setminus \tilde{S}(\mathbf{o}, \mathbf{r}, \epsilon_1) \rightarrow \mathbb{R}^+, \phi(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) = \frac{\bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \bar{J}^*(\mathbf{o}; \mathbf{r})}{\min_i \bar{c}_i(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})}$$

Section 5: Construction of the Loss function for Training

3. Continuity issue: first step

Theorem 2 (with Assumption 1)

For any positive ϵ_1 , there exists $\underline{\beta}(\mathbf{o}, \mathbf{r}, \epsilon_1) > 0$ s.t., any $\beta > \underline{\beta}$ makes all solutions of the converted problem is at most at a distance ϵ_1 from a solution of the original problem with the same $\mathbf{z}$.

$$\forall \beta > \underline{\beta}, \forall (\tilde{\mathbf{x}}, \mathbf{z}) \in \arg \min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \beta^\top \min(\bar{\mathbf{c}}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), \mathbf{0}),$$
$$\exists \mathbf{x}^*, \|\mathbf{x}^* - \tilde{\mathbf{x}}\| < \epsilon_1, \mathbf{x}^* \in S_{\mathbf{z}}(\mathbf{o}, \mathbf{r})$$

Section 5: Construction of the Loss function for Training

3. Continuity issue: first step

Theorem 2 (with Assumption 1)

For any positive ϵ_1 , there exists $\underline{\beta}(\mathbf{o}, \mathbf{r}, \epsilon_1) > 0$ s.t., any $\beta > \underline{\beta}$ makes all solutions of the converted problem is at most at a distance ϵ_1 from a solution of the original problem with the same $\mathbf{z}$.

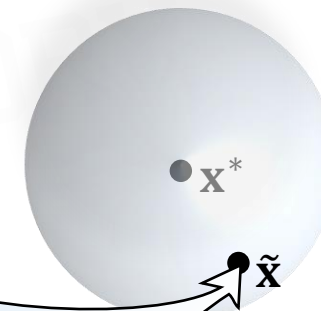
explain 6:

Original optimization problem

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{X}, \mathbf{z} \in \mathcal{Z}} \quad & \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \\ \text{subject to} \quad & \bar{c}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) \geq 0 \end{aligned}$$

Converted optimization problem

$$\min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \beta^\top \min(\bar{c}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), 0)$$



$$\beta > \underline{\beta}(\mathbf{o}, \mathbf{r}, \epsilon_1)$$

Section 5: Construction of the Loss function for Training

4. Continuity issue: second step

Assumption 2

- a) $\bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{o}$.
- b) $\bar{c}_i(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{o}$.
- c) $L(\mathbf{x}, \mathbf{z}, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$ is continuous with respect to $\mathbf{x}$ and $\mathbf{o}$.

Section 5: Construction of the Loss function for Training

4. Continuity issue: second step

Assumption 2

- a) $\bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{o}$.
- b) $\bar{c}_i(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{o}$.
- c) $L(\mathbf{x}, \mathbf{z}, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$ is continuous with respect to $\mathbf{x}$ and $\mathbf{o}$.

explain 1: objective function is continuous w.r.t the parameters. (Continuity issue - b)

Section 5: Construction of the Loss function for Training

4. Continuity issue: second step

Assumption 2

- a) $\bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{o}$.
- b) $\bar{c}_i(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r})$ is continuous with respect to $\mathbf{o}$.
- c) $L(\mathbf{x}, \mathbf{z}, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$ is continuous with respect to $\mathbf{x}$ and $\mathbf{o}$.

explain 3: different from the continuity of $L(\mathbf{x}^*(\mathbf{o}, \mathbf{r}), \mathbf{z}^*(\mathbf{o}, \mathbf{r}), \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$.

$\rightarrow L^*(\mathbf{o}; \boldsymbol{\theta}, \mathbf{r}) := \min_{(\mathbf{x}, \mathbf{z}) \in S(\mathbf{o}, \mathbf{r})} L(\mathbf{x}, \mathbf{z}, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$ *well defined!*

$\rightarrow$ *Now, we want to show the convergence of $\tilde{L}$ to L^* in Theorem 3-b.*

explain 4: also support Continuity issue – a,b

Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

1) Approximated loss function construction (with the converted optimization problem)

$$\begin{aligned}\tilde{L}(\mathbf{o}; \boldsymbol{\theta}, \mathbf{r}, \boldsymbol{\beta}, \lambda) = & \frac{1}{\lambda} \left(\min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} (\lambda L(\mathbf{x}, \mathbf{z}, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r}) + \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \boldsymbol{\beta}^\top \min(\bar{\mathbf{c}}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), \mathbf{0})) \right. \\ & \left. - \min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} (\bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \boldsymbol{\beta}^\top \min(\bar{\mathbf{c}}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), \mathbf{0})) \right)\end{aligned}$$

Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

1) Approximated loss function construction (with the converted optimization problem)

$$\tilde{L}(\mathbf{o}; \boldsymbol{\theta}, \mathbf{r}, \boldsymbol{\beta}, \lambda) = \frac{1}{\lambda} \left(\min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} (\lambda L(\mathbf{x}, \mathbf{z}, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r}) + \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \boldsymbol{\beta}^\top \min(\bar{\mathbf{c}}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), \mathbf{0})) \right. \\ \left. - \min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} (\bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \boldsymbol{\beta}^\top \min(\bar{\mathbf{c}}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), \mathbf{0})) \right)$$

Remind 1: Continuity issue

- a) objective function ~~and constraints~~ are continuous w.r.t the variables.
- b) objective function is continuous w.r.t the parameters.
- c) constraints are independent of the parameters.

Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

1) Approximated loss function construction (with the converted optimization problem)

$$\begin{aligned}\tilde{L}(\mathbf{o}; \boldsymbol{\theta}, \mathbf{r}, \beta, \lambda) = & \frac{1}{\lambda} \left(\min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} (\lambda L(\mathbf{x}, \mathbf{z}, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r}) + \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \beta^\top \min(\bar{\mathbf{c}}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), \mathbf{0})) \right. \\ & \left. - \min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} (\bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \beta^\top \min(\bar{\mathbf{c}}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), \mathbf{0})) \right)\end{aligned}$$

Remind 2: Alternative of the loss function

- approximated general loss function: $\tilde{L} \simeq L(\mathbf{x}^*(\mathbf{o}, \mathbf{r}), \mathbf{z}^*(\mathbf{o}, \mathbf{r}), \mathbf{o}; \boldsymbol{\theta}, \mathbf{r})$ (Theorem 3-b)
- continuous with respect of the network output. (Theorem 3-a, already check!)
- well defined.
- avoid two-stage minimization.

Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

Theorem 3. (with Assumption 1 & 2)

For any $(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r})$ and $\epsilon_2 > 0$, two following statements hold.

- a) $\tilde{L}(\mathbf{o}; \boldsymbol{\theta}, \mathbf{r}, \beta, \lambda)$ is continuous w.r.t $\mathbf{o}$ for any $\beta(> 0)$ and $\lambda(> 0)$.
- b) We can find $(\beta_0(\mathbf{o}, \mathbf{r}, \lambda, \boldsymbol{\theta}, \epsilon_2), \lambda_0(\mathbf{o}, \mathbf{r}, \boldsymbol{\theta}, \epsilon_2))$ that makes any $(\beta > \beta_0(\mathbf{o}, \mathbf{r}, \lambda, \boldsymbol{\theta}, \epsilon_2), \lambda < \lambda_0(\mathbf{o}, \mathbf{r}, \boldsymbol{\theta}, \epsilon_2))$ guarantees $|\tilde{L}(\mathbf{o}; \boldsymbol{\theta}, \mathbf{r}, \beta, \lambda) - L^*(\mathbf{o}; \boldsymbol{\theta}, \mathbf{r})| < \epsilon_2$.

Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

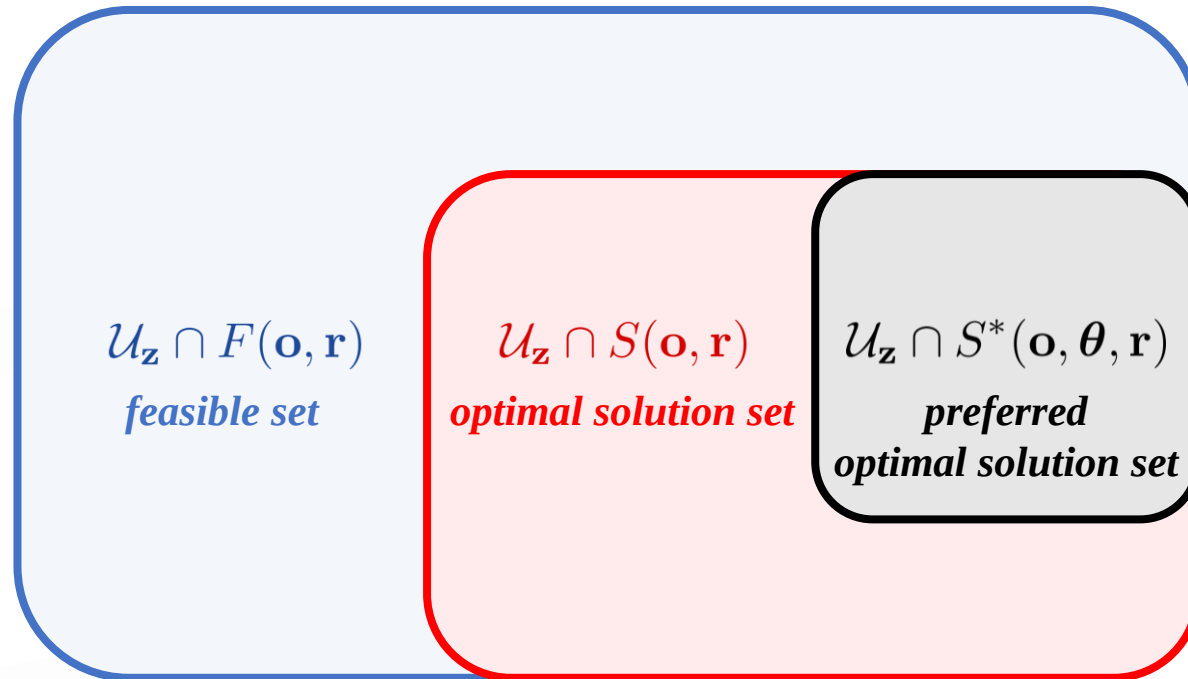
2) Proof of Theorem 3-b: Shortcut

- Step 1: find λ_0 s.t. for all $\lambda < \lambda_0$, $|\lambda L(\mathbf{x}^*, \mathbf{z}^*, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r}) + \bar{J}(\mathbf{x}^*, \mathbf{z}^*; \mathbf{o}, \mathbf{r}) - \lambda L^*(\mathbf{o}; \boldsymbol{\theta}, \mathbf{r}) - \bar{J}^*(\mathbf{o}; \mathbf{r})| < \frac{\lambda \epsilon_2}{3}$.
- Step 2: shortcut.

Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

3) Proof of Theorem 3-b: Notation



optimal solution set

$$(1) \min_{\mathbf{x}, \mathbf{z}} \bar{J} \quad \text{s.t. } \bar{c} \geq 0$$

preferred optimal solution set

$$(1) \min_{\mathbf{x}, \mathbf{z}} \bar{J} \quad \text{s.t. } \bar{c} \geq 0$$
$$(6) \min_{\mathbf{x}, \mathbf{z}} L$$

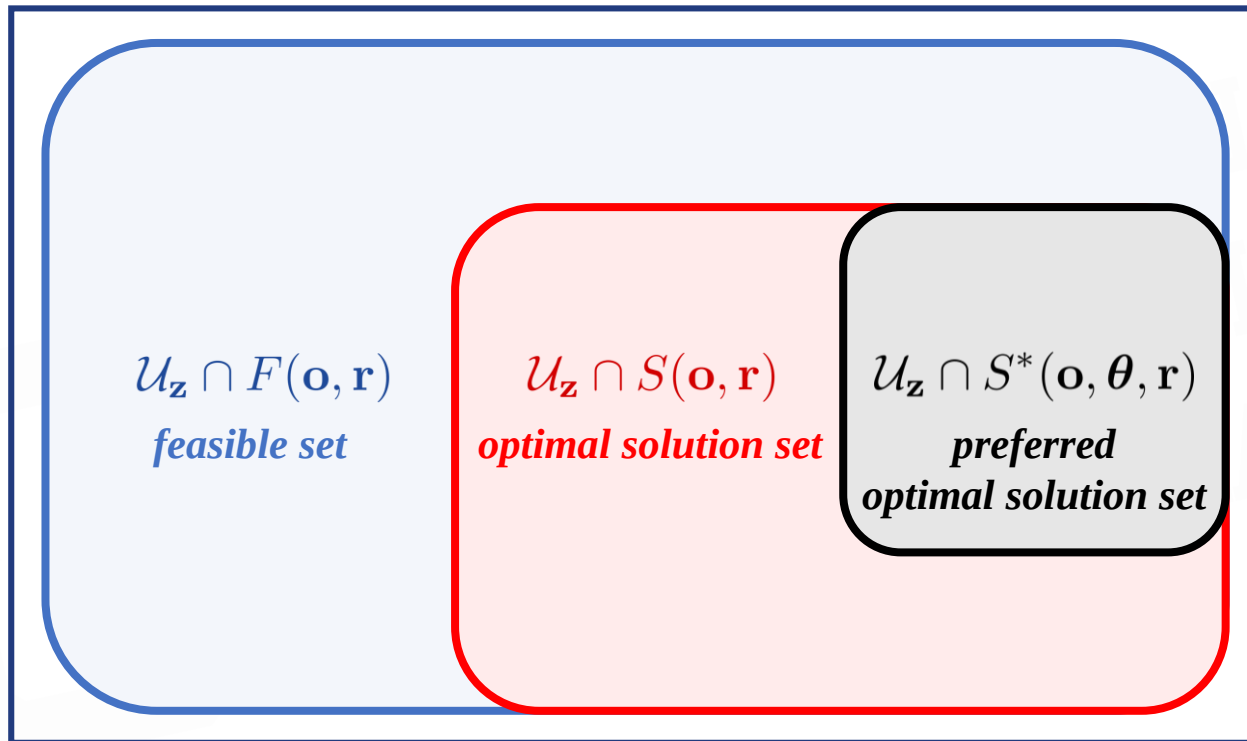
*solution set of
the integrated optimization layer*

$$(7) \min_{\mathbf{x}, \mathbf{z}} \lambda L + \bar{J} \quad \text{s.t. } \bar{c} \geq 0$$

Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

3) Proof of Theorem 3-b: Notation



explain 1: all region is compact. (intersection of compact & closed)

explain 2: L is uniform continuous in all region.

optimal solution set

$$(1) \min_{\mathbf{x}, \mathbf{z}} \bar{J} \quad \text{s.t. } \bar{c} \geq 0$$

preferred optimal solution set

$$(1) \min_{\mathbf{x}, \mathbf{z}} \bar{J} \quad \text{s.t. } \bar{c} \geq 0$$
$$(6) \min_{\mathbf{x}, \mathbf{z}} L$$

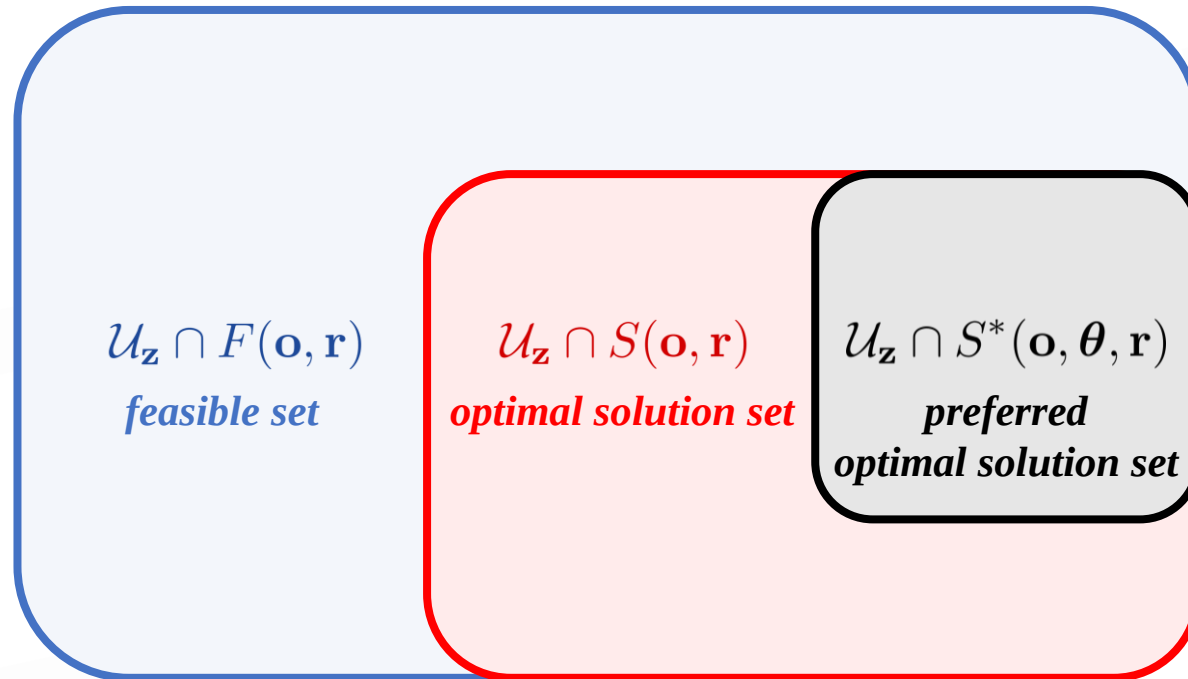
*solution set of
the integrated optimization layer*

$$(7) \min_{\mathbf{x}, \mathbf{z}} \lambda L + \bar{J} \quad \text{s.t. } \bar{c} \geq 0$$

Section 5: Construction of the Loss function for Training

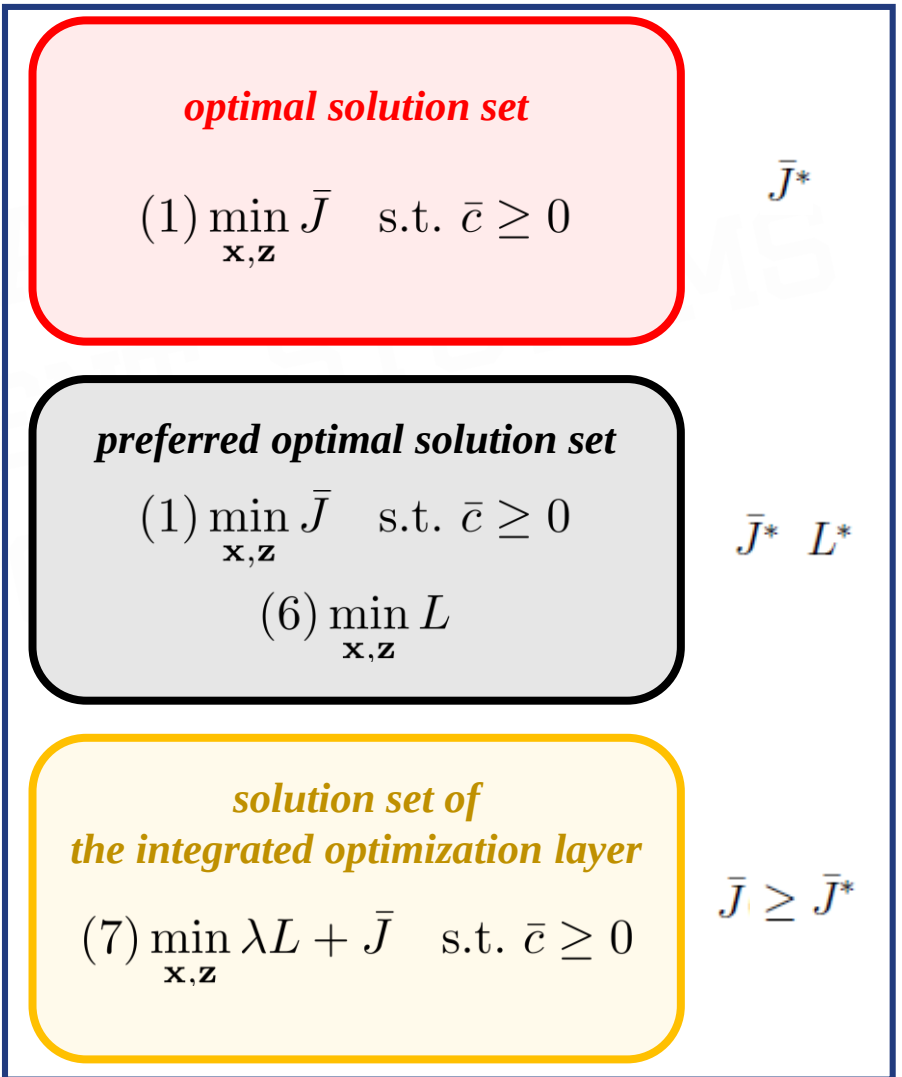
5. Approximated loss function: convergence to real loss

3) Proof of Theorem 3-b: Notation



explain 3: same feasible set.

explain 4: set inclusion.

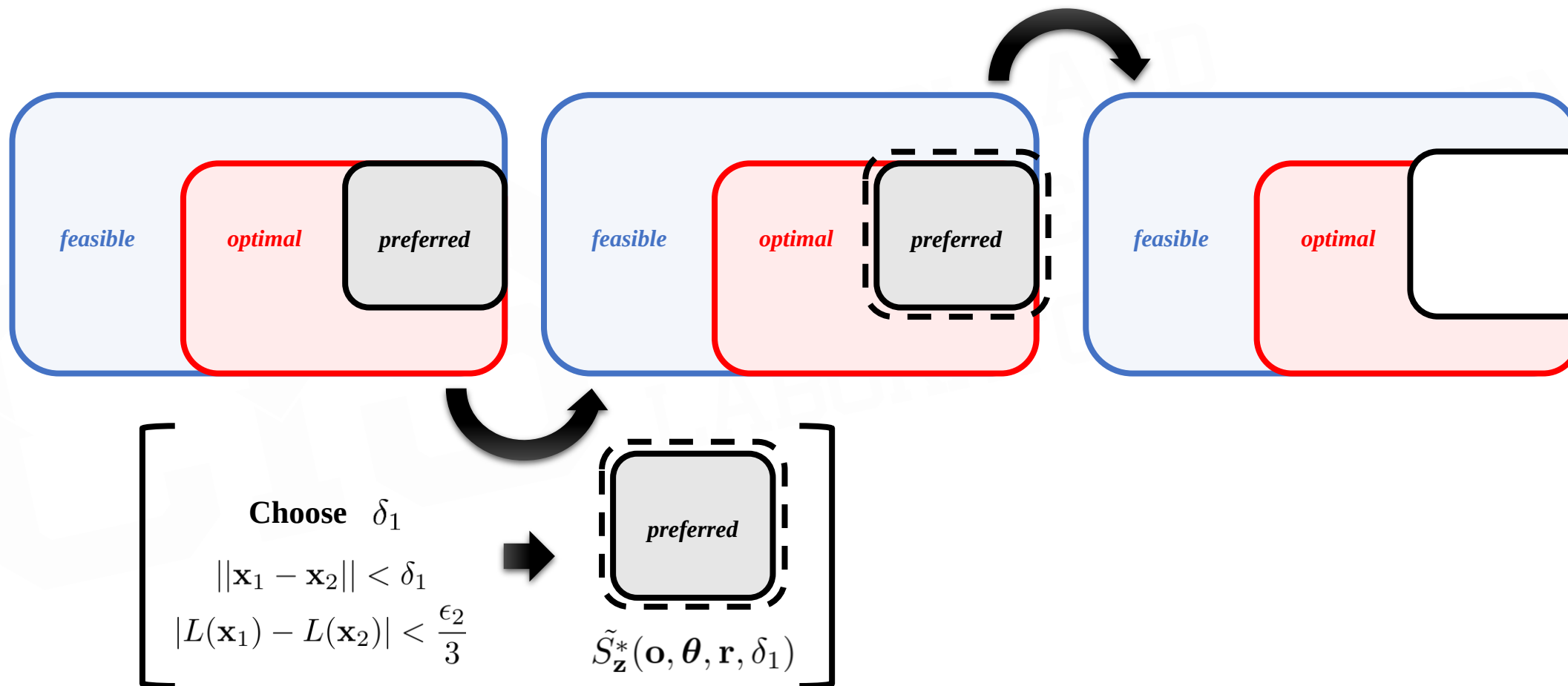


Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

4) Proof of Theorem 3-b: Step 1

$$(\mathcal{U}_{\mathbf{z}} \cap S(\mathbf{o}, \mathbf{r})) \setminus \tilde{S}_{\mathbf{z}}^*(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \delta_1)$$

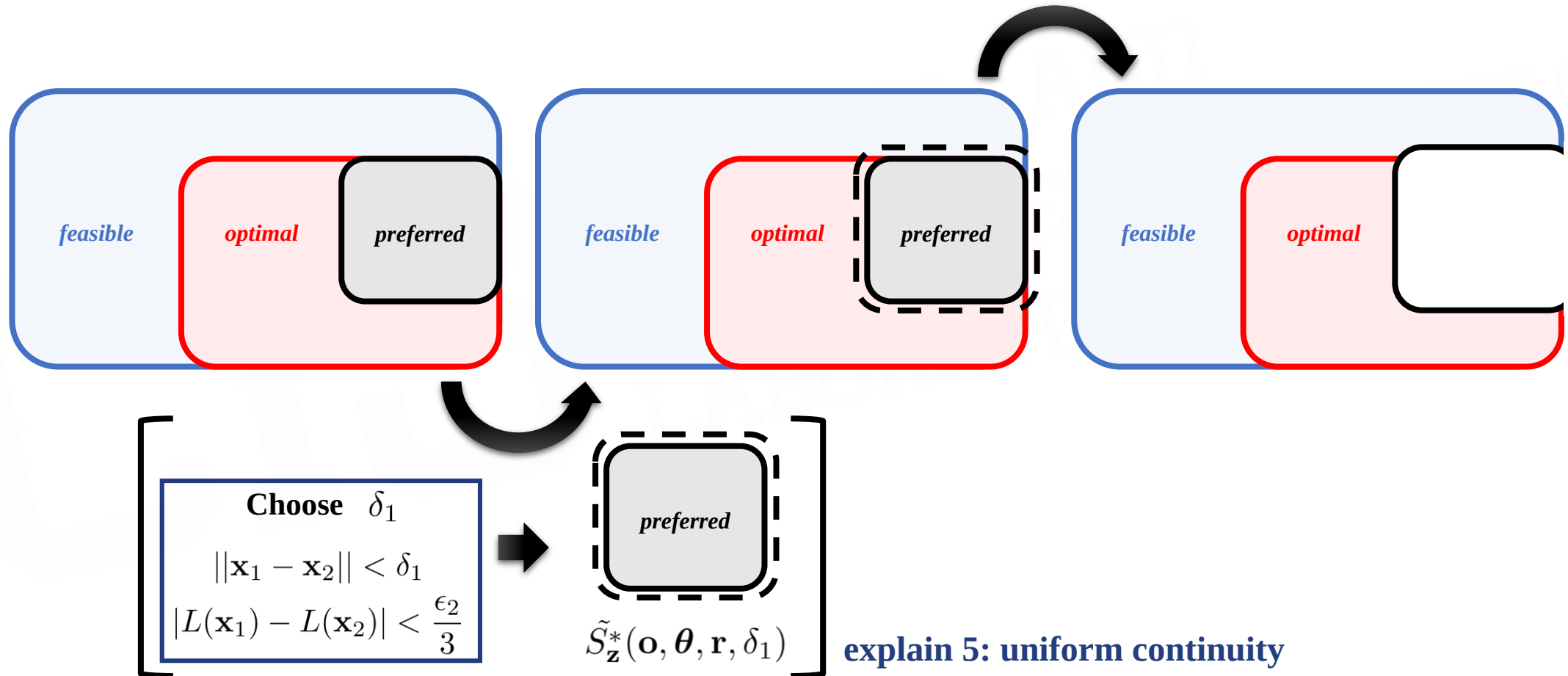


Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

4) Proof of Theorem 3-b: Step 1

$$(\mathcal{U}_{\mathbf{z}} \cap S(\mathbf{o}, \mathbf{r})) \setminus \tilde{S}_{\mathbf{z}}^*(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \delta_1)$$

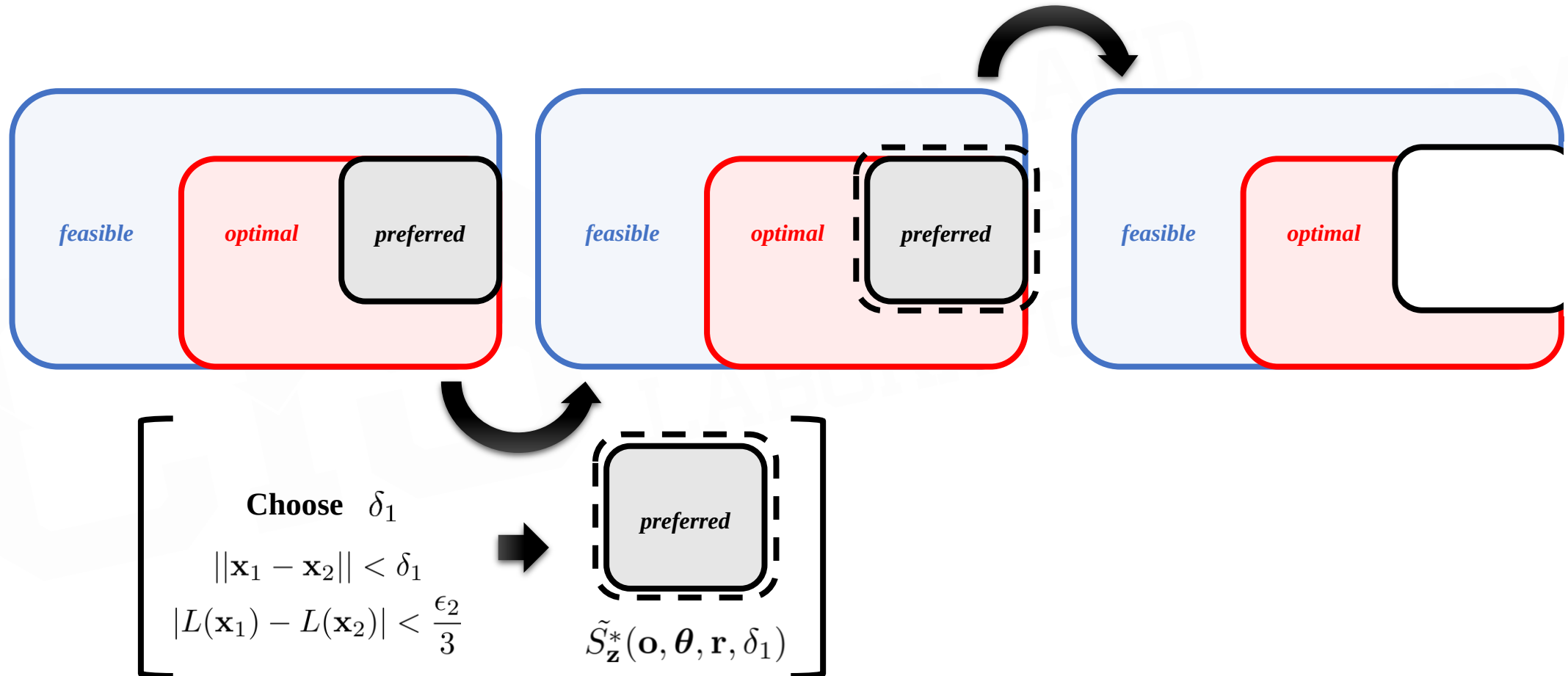


Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

4) Proof of Theorem 3-b: Step 1

$(\mathcal{U}_{\mathbf{z}} \cap S(\mathbf{o}, \mathbf{r})) \setminus \tilde{S}_{\mathbf{z}}^*(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \delta_1)$ **explain 6: compact**

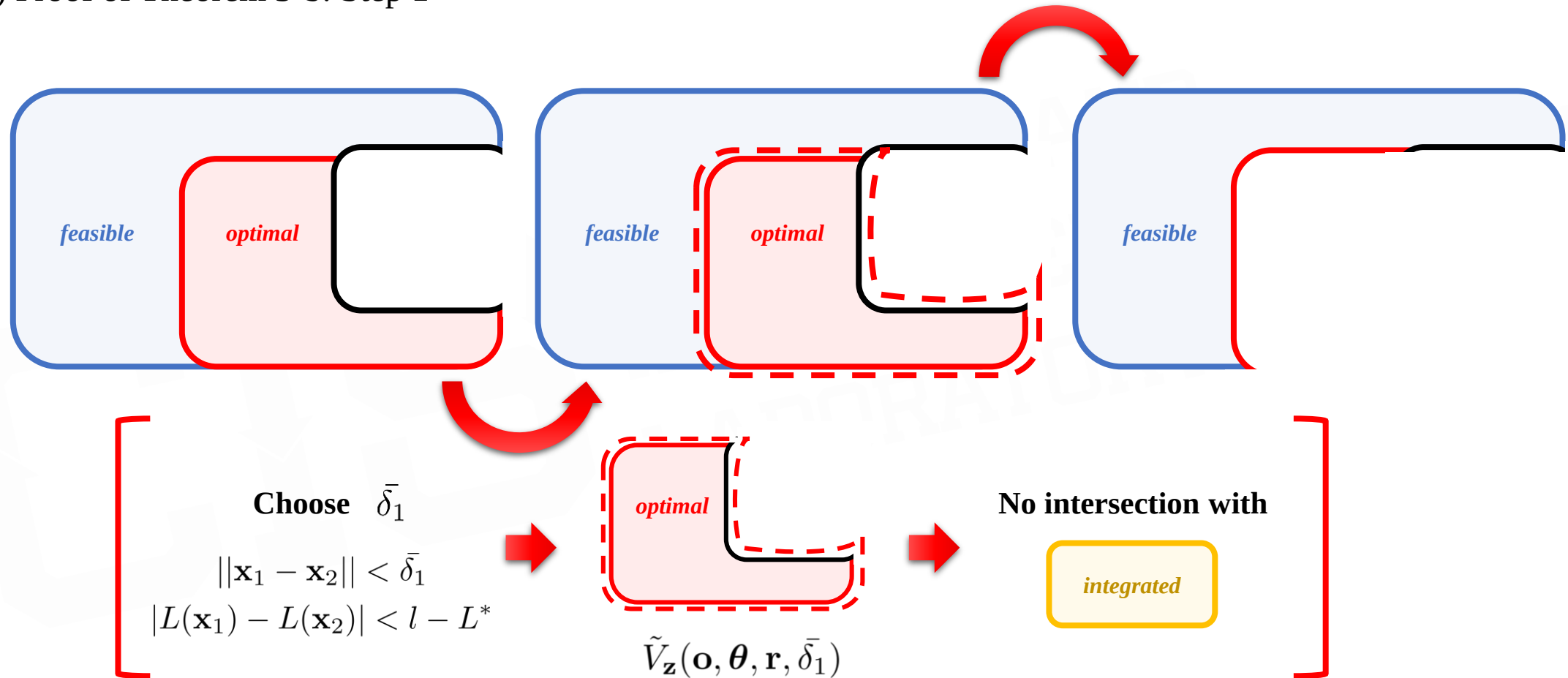


Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

$$((\mathcal{U}_{\mathbf{z}} \cap F(\mathbf{o}, \mathbf{r})) \setminus \tilde{S}_{\mathbf{z}}^*(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \delta_1)) \setminus \tilde{V}_{\mathbf{z}}(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \bar{\delta}_1)$$

4) Proof of Theorem 3-b: Step 1

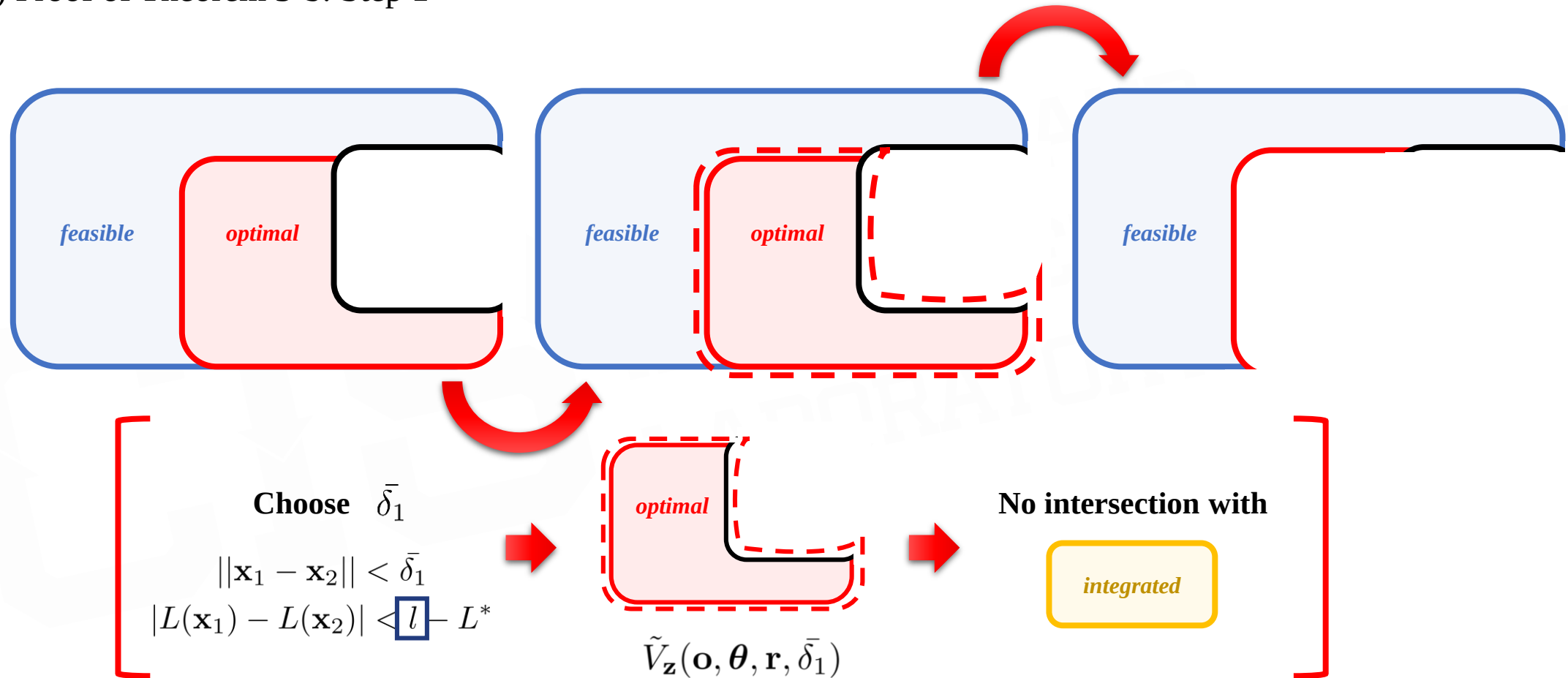


Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

$$((\mathcal{U}_{\mathbf{z}} \cap F(\mathbf{o}, \mathbf{r})) \setminus \tilde{S}_{\mathbf{z}}^*(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \delta_1)) \setminus \tilde{V}_{\mathbf{z}}(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \bar{\delta}_1)$$

4) Proof of Theorem 3-b: Step 1



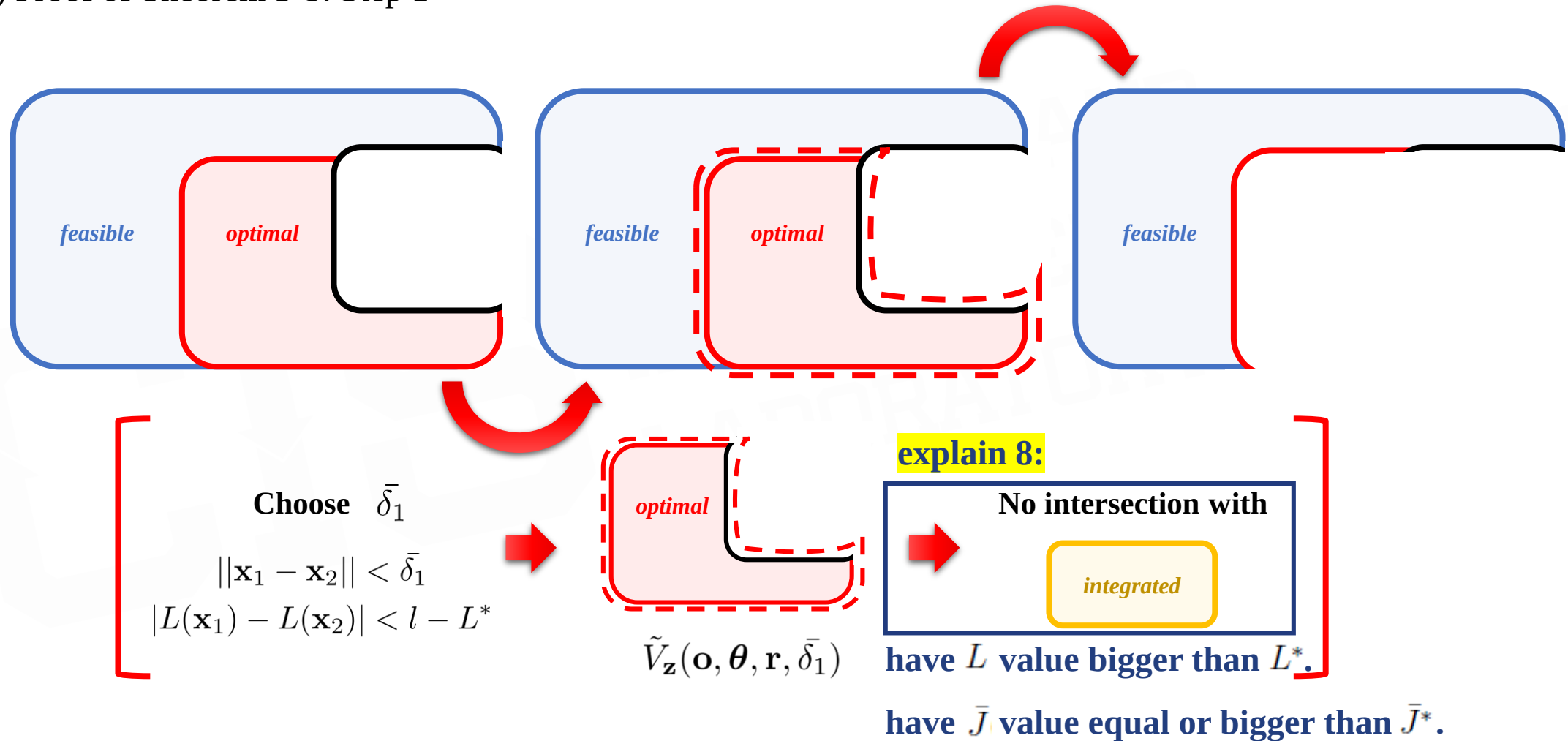
explain 7: minimum of red section. (compact + continuous)

Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

$$((\mathcal{U}_z \cap F(\mathbf{o}, \mathbf{r})) \setminus \tilde{S}_z^*(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \delta_1)) \setminus \tilde{V}_z(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \bar{\delta}_1)$$

4) Proof of Theorem 3-b: Step 1

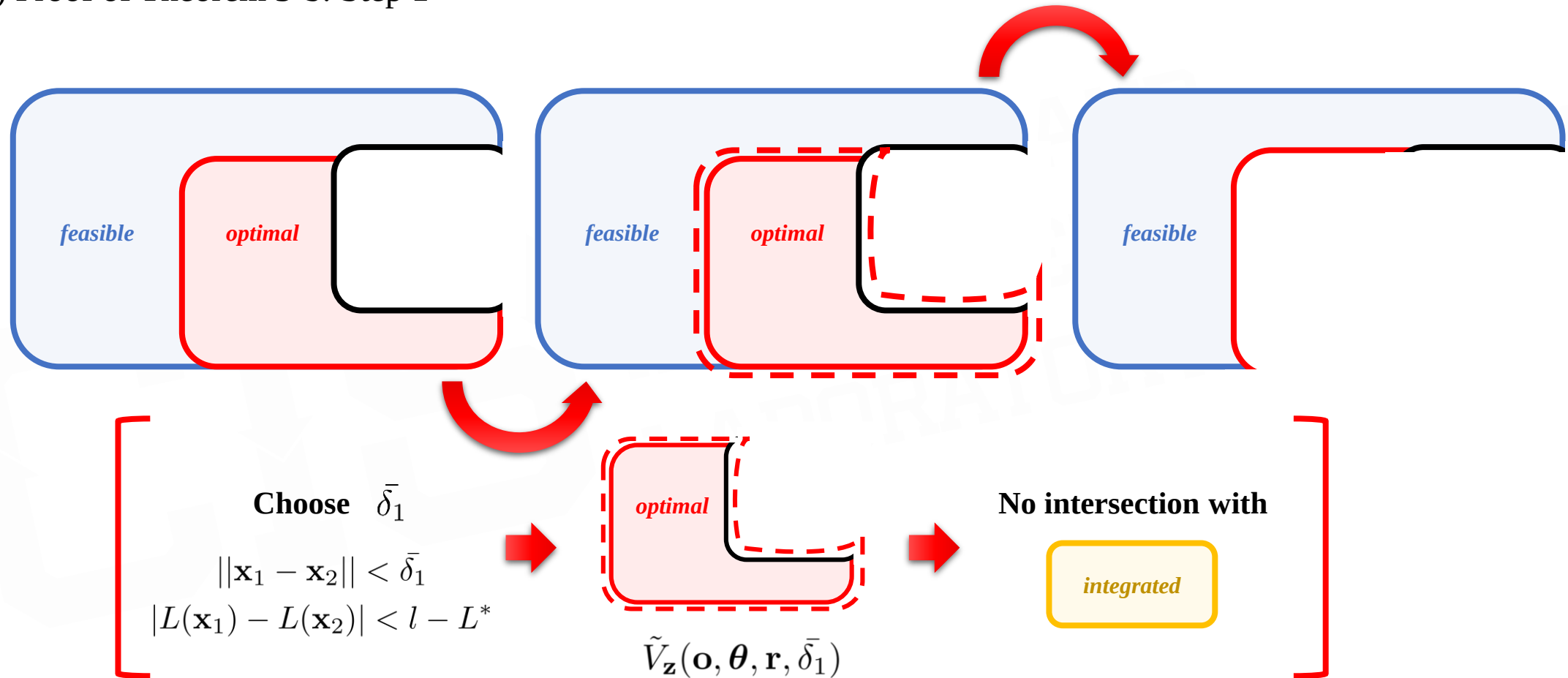


Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

4) Proof of Theorem 3-b: Step 1

$$((\mathcal{U}_{\mathbf{z}} \cap F(\mathbf{o}, \mathbf{r})) \setminus \tilde{S}_{\mathbf{z}}^*(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \delta_1)) \setminus \tilde{V}_{\mathbf{z}}(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \bar{\delta}_1) \quad \text{explain 9: compact}$$



Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

4) Proof of Theorem 3-b: Step 1

Any optimal solution $(\mathbf{x}^*, \mathbf{z}^*)$ of the integrated optimization problem satisfy

$$|\lambda L + \bar{J} - \lambda L^* - \bar{J}^*| < \frac{\lambda \epsilon_2}{3}$$

$$\tilde{S}_{\mathbf{z}}^*(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \delta_1)$$

Step1 Conclusion

For any positive $\lambda < \underline{\lambda}$,
no intersection with

integrated

feasible

$$\begin{aligned} q_0 &: \text{the minimum values of } \bar{J} \\ \bar{l} &: \text{the maximum values of } L \\ \underline{l} &: \text{the minimum values of } L \end{aligned} \quad \begin{aligned} \underline{\lambda}_z &:= \frac{q_0 - \bar{J}^*}{\bar{l} - \underline{l}} \\ \underline{\lambda} &:= \min_{\mathbf{z} \in Proj_{\mathcal{Z}}(\mathcal{U})} \underline{\lambda}_z \end{aligned}$$

Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

4) Proof of Theorem 3-b: Step 1

Any optimal solution $(\mathbf{x}^*, \mathbf{z}^*)$ of the integrated optimization problem satisfy

$$|\lambda L + \bar{J} - \lambda L^* - \bar{J}^*| < \frac{\lambda \epsilon_2}{3}$$

$$\tilde{S}_{\mathbf{z}}^*(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \delta_1)$$

Step1 Conclusion

For any positive $\lambda < \underline{\lambda}$,
no intersection with

integrated

feasible

q_0 : the minimum values of $\bar{J}$
 $\bar{l}$: the maximum values of L
 $\underline{l}$: the minimum values of L

$$\underline{\lambda}_z := \frac{q_0 - \bar{J}^*}{\bar{l} - \underline{l}}$$

$$\underline{\lambda} := \min_{\mathbf{z} \in Proj_{\mathcal{Z}}(\mathcal{U})} \underline{\lambda}_z$$

explain 10: compact + continuous

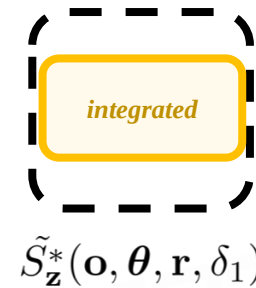
Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

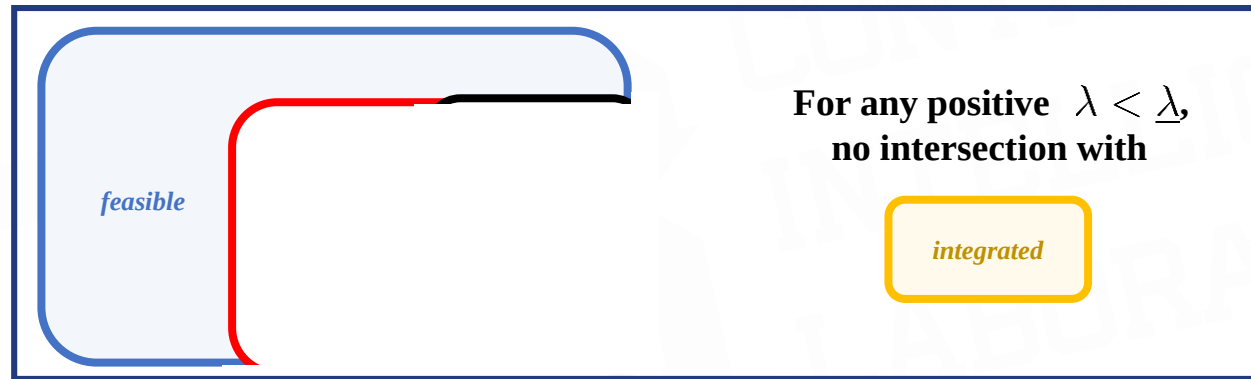
4) Proof of Theorem 3-b: Step 1

Any optimal solution $(\mathbf{x}^*, \mathbf{z}^*)$ of the integrated optimization problem satisfy

$$|\lambda L + \bar{J} - \lambda L^* - \bar{J}^*| < \frac{\lambda \epsilon_2}{3}$$



Step1 Conclusion



explain 11:

have $\bar{J}$ value bigger than $\bar{J}^*$.

can have L value equal or smaller than L^* .

q_0 : the minimum values of $\bar{J}$
 $\bar{l}$: the maximum values of L
 $\underline{l}$: the minimum values of L

$$\underline{\lambda}_z := \frac{q_0 - \bar{J}^*}{\bar{l} - \underline{l}}$$

$$\underline{\lambda} := \min_{\mathbf{z} \in Proj_{\mathcal{Z}}(\mathcal{U})} \underline{\lambda}_z$$

Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real explain 12: details are mentioned in equation (32,33). about $\bar{J}$.

4) Proof of Theorem 3-b: Step 1

Any optimal solution $(\mathbf{x}^*, \mathbf{z}^*)$ of the integrated optimization problem satisfy

$$|\lambda L + \bar{J} - \lambda L^* - \bar{J}^*| < \frac{\lambda \epsilon_2}{3}$$

$$\tilde{S}_{\mathbf{z}}^*(\mathbf{o}, \boldsymbol{\theta}, \mathbf{r}, \delta_1)$$

Step1 Conclusion

For any positive $\lambda < \underline{\lambda}$,
no intersection with

integrated

feasible

q_0 : the minimum values of $\bar{J}$
 $\bar{l}$: the maximum values of L
 $\underline{l}$: the minimum values of L

$$\underline{\lambda}_z := \frac{q_0 - \bar{J}^*}{\bar{l} - \underline{l}}$$

$$\underline{\lambda} := \min_{\mathbf{z} \in Proj_{\mathcal{Z}}(\mathcal{U})} \underline{\lambda}_z$$

Section 5: Construction of the Loss function for Training

5. Approximated loss function: convergence to real loss

3) Proof of Theorem 3-b: Step 2 shortcut

(1), (2) fixed λ and find β_0 !

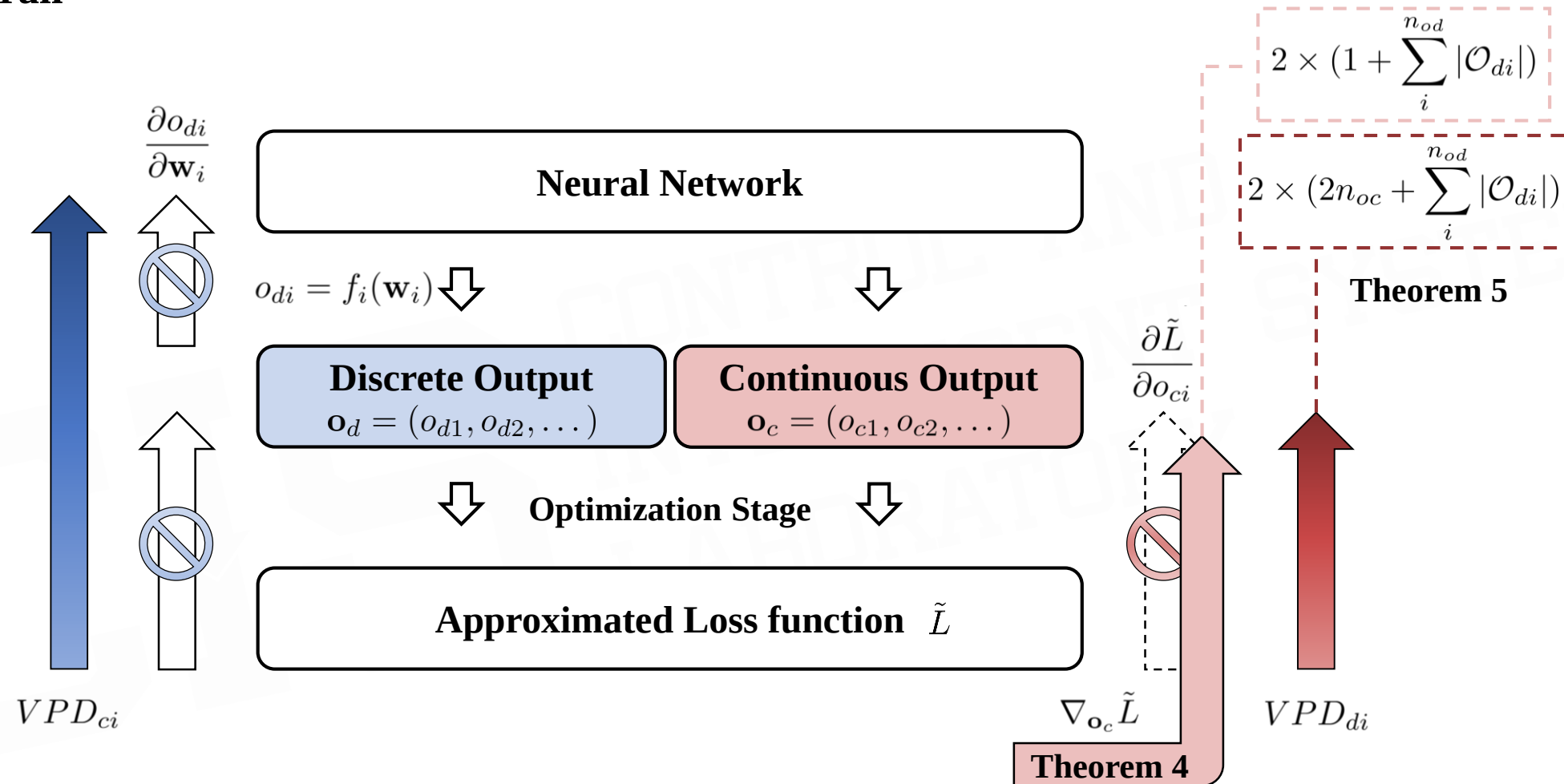
$$\lambda|\tilde{L}(\mathbf{o}; \boldsymbol{\theta}, \mathbf{r}, \beta, \lambda) - L^*(\mathbf{o}; \boldsymbol{\theta}, \mathbf{r})| \leq \begin{array}{l} \text{(1)} \quad \left| \min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} (\lambda L(\mathbf{x}, \mathbf{z}, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r}) + \bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \beta^\top \min(\bar{\mathbf{c}}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), 0)) \right. \\ \quad \left. - \lambda L(\mathbf{x}^*, \mathbf{z}^*, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r}) - \bar{J}(\mathbf{x}^*, \mathbf{z}^*; \mathbf{o}, \mathbf{r}) \right| \\ \text{(2)} \quad + \left| \min_{(\mathbf{x}, \mathbf{z}) \in \mathcal{U}} (\bar{J}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}) - \beta^\top \min(\bar{\mathbf{c}}(\mathbf{x}, \mathbf{z}; \mathbf{o}, \mathbf{r}), 0)) - \bar{J}^*(\mathbf{o}; \mathbf{r}) \right| \\ \text{(3)} \quad + \left| \lambda L(\mathbf{x}^*, \mathbf{z}^*, \mathbf{o}; \boldsymbol{\theta}, \mathbf{r}) + \bar{J}(\mathbf{x}^*, \mathbf{z}^*; \mathbf{o}, \mathbf{r}) - \lambda L^*(\mathbf{o}; \boldsymbol{\theta}, \mathbf{r}) - \bar{J}^*(\mathbf{o}; \mathbf{r}) \right| \end{array} < \frac{\lambda \epsilon_2}{3} + \frac{\lambda \epsilon_2}{3} + \frac{\lambda \epsilon_2}{3} = \lambda \epsilon_2$$

(3) already did it.

Section 6: Calculation of gradient (brief)

Section 6: Calculation of gradient (brief)

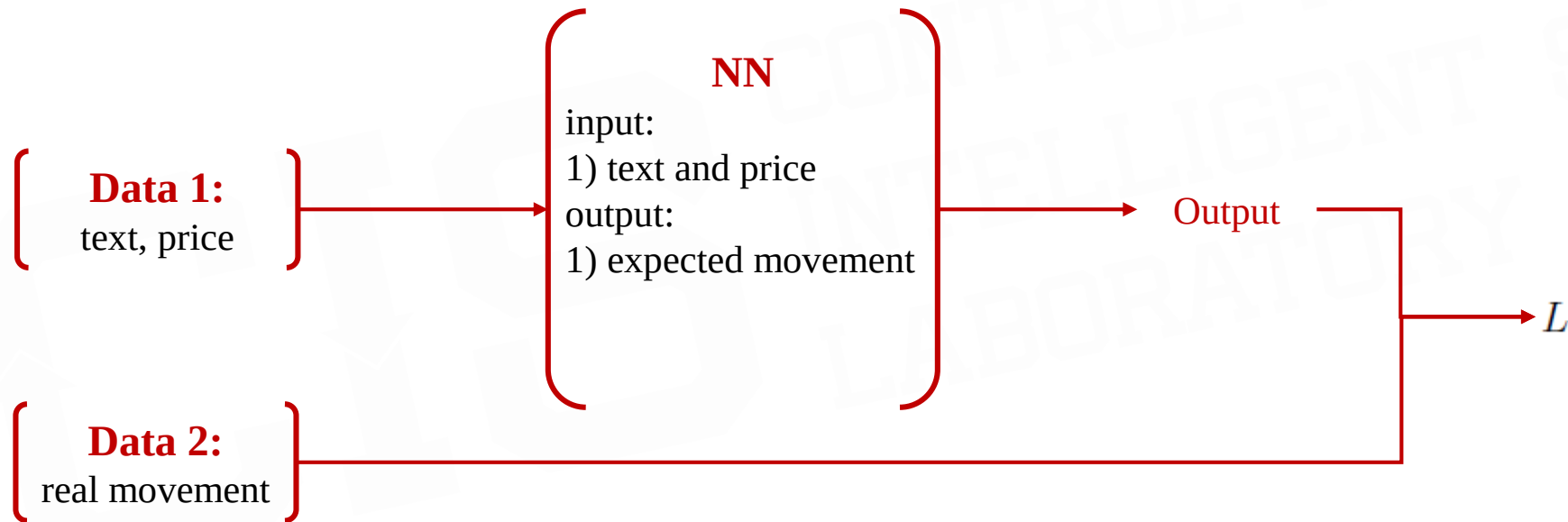
1. Overall



Section 7: Application 1: Stock trading

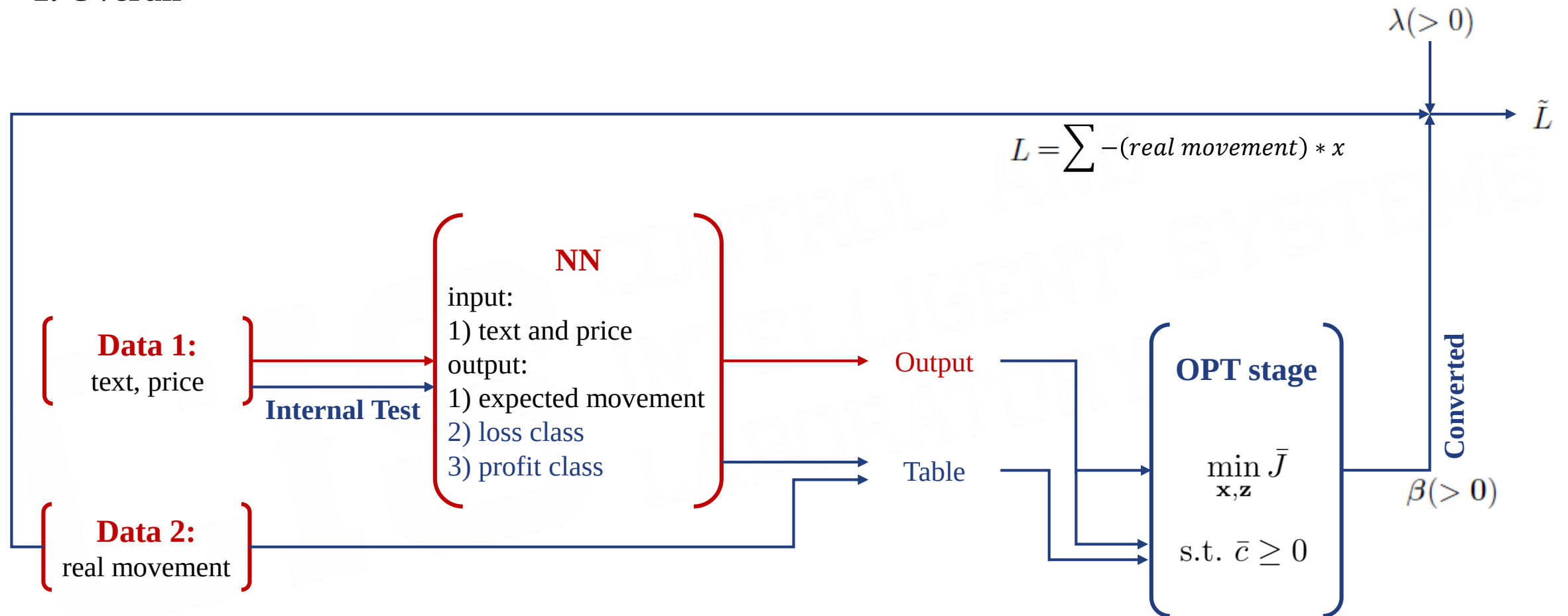
Section 7: Application 1: Stock trading

1. Overall



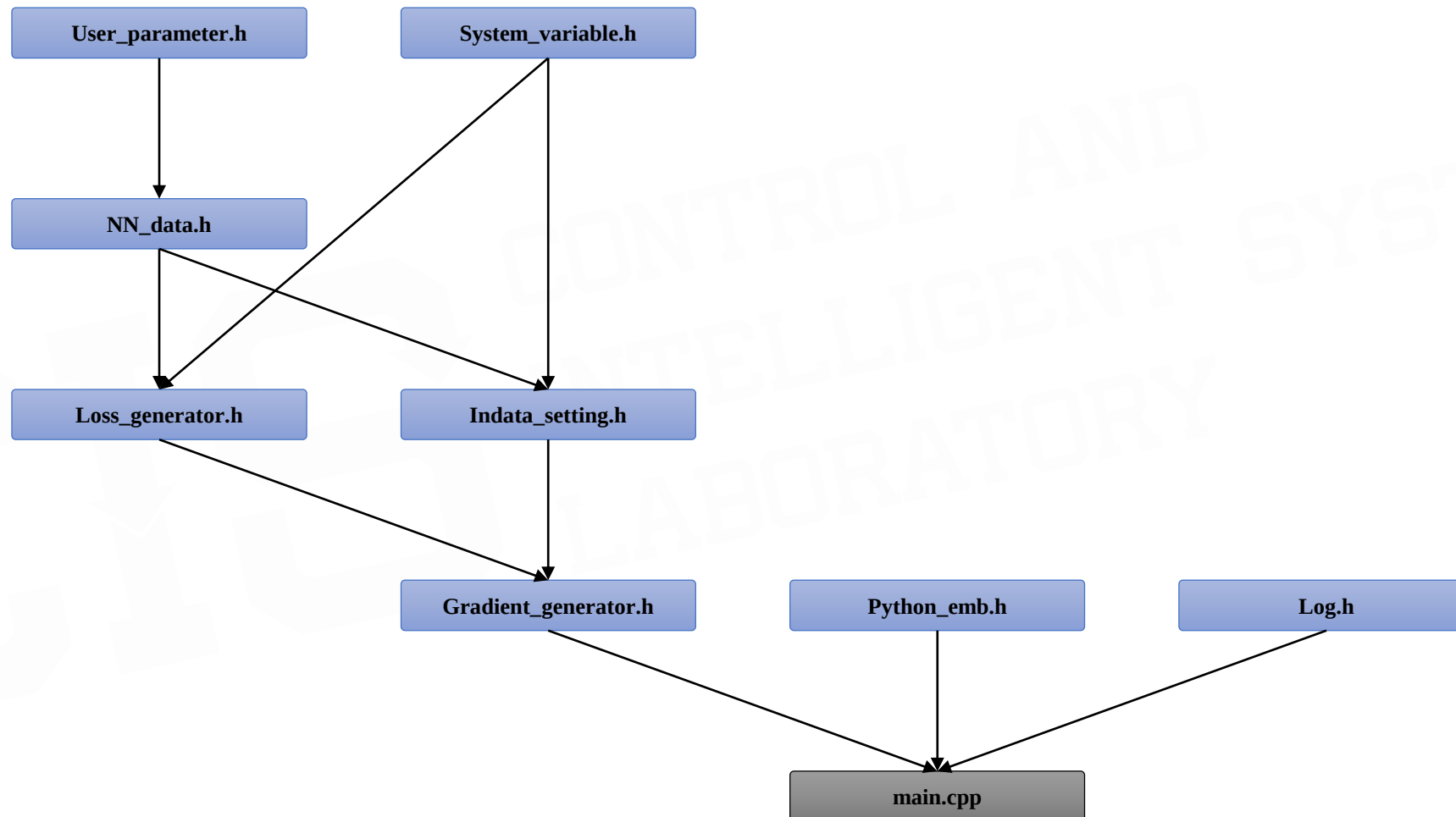
Section 7: Application 1: Stock trading

1. Overall



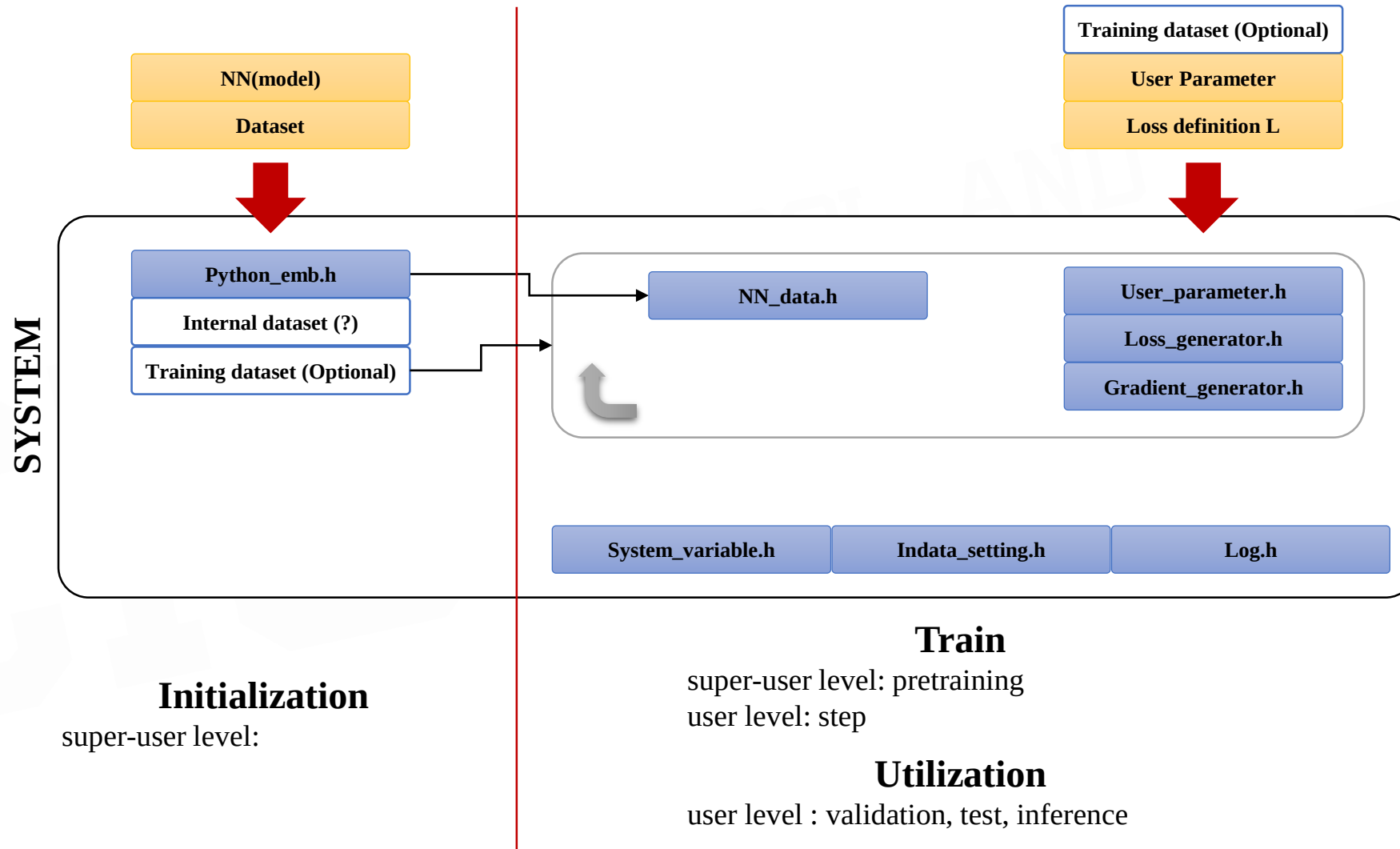
Section 7: Application 1: Stock trading

2. System structure rebuilding (in progress)



Section 7: Application 1: Stock trading

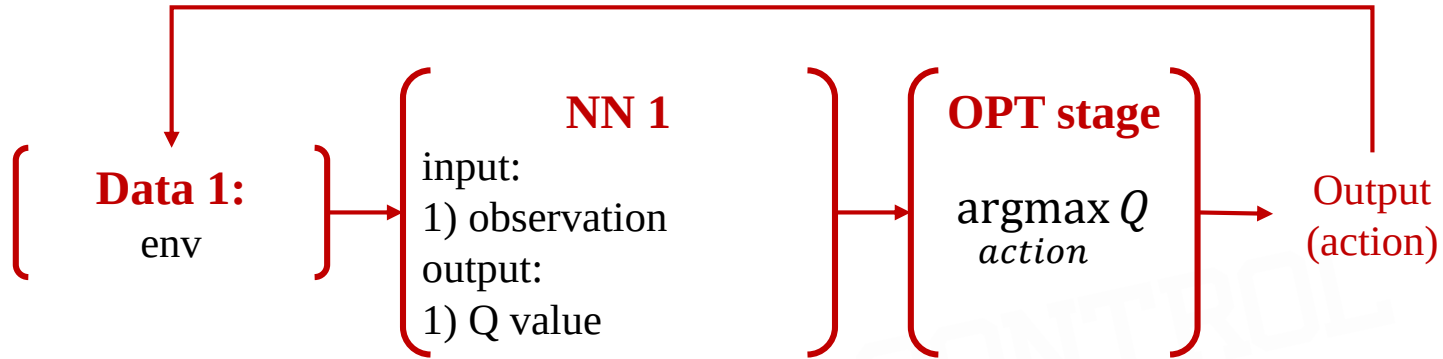
2. System structure rebuilding (in progress)



Section 8: Application 2: Safety gym

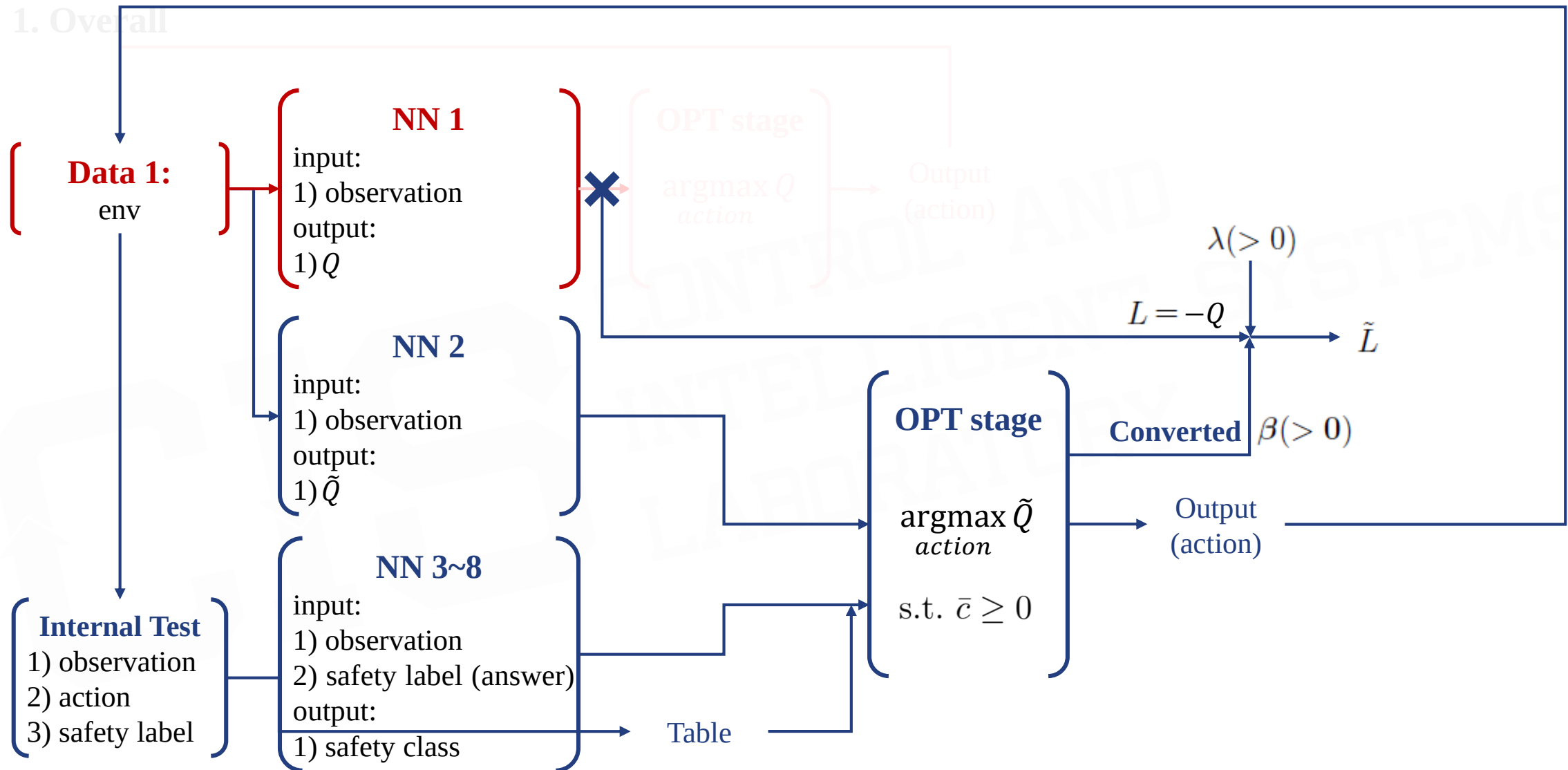
Section 8: Application 2: Safety gym

1. Overall



Section 8: Application 2: Safety gym

1. Overall



A General User-Customizable Predict+Optimize Framework

(Supplementary)

23.10.06

Kangyeon Kim